

Do AI weather models miss extremes?

Evidence from ten months of verification against European weather stations

Jua Team*

July 2026

Abstract

First-generation AI weather models are often reported to underperform at extremes, mostly in reanalysis-based evaluations of deterministic regression systems. We verify eleven physical and AI forecast systems against European synoptic, solar and rain-gauge stations over ten months for 10 m wind, 2 m temperature, hourly shortwave accumulation and hourly precipitation, scoring MAE against ECMWF IFS in ERA5 1991–2020 climatological regimes. Among these systems, AI models do not show a uniform relative-skill deficit in the tails. Jua EPT-2.1 Europa leads all-conditions wind (+8.4%), while Jua EPT-2 HRRR leads temperature overall (+12.1%) and in the heat regime (+19.6 ± 2.2%). EPT-2.1 Europa and DWD ICON Global lead at gale-force wind. Jua EPT-2.1 Helios leads solar overall (+10.2 ± 1.7%), in overcast conditions (+16.4 ± 3.4%), and in the clear-sky tail (+24.8 ± 5.4%). For precipitation, three Jua models gain 14–15% at moderate intensity and 9–11% at P75–P95; EPT-2 Reasoning remains ahead at >P95 (+1.7 ± 0.5%). Failures are model-specific: ECMWF AIFS loses 4.9 ± 2.0% in the heat tail, while NOAA GFS loses 22.8 ± 2.0% there. Every model, NWP included, shows a shared conditional bias toward the centre of the observed distribution, with an inter-model spread several times smaller than the shared signal. Missing relative skill at extremes is therefore not a property of AI weather models as a class, but of particular AI and physical models.

1 Introduction

Data-driven weather models outperform operational physical systems on standard forecast scores [Bi et al., 2023, Lam et al., 2023, Ben Bouallègue et al., 2024]. A remaining concern is extremes: models trained with mean squared error are expected to regress toward the conditional mean, smooth sharp gradients, and underforecast rare events [Olivetti and Messori, 2024a, Bonavita, 2024]. Case studies and systematic evaluations report supporting evidence, including underestimated peak winds in storm Ciarán [Charlton-Perez et al., 2024], underestimated hot and windy tails at longer leads [Olivetti and Messori, 2024b], and physical models ahead of AI on record-breaking events [Zhang et al., 2026].

Most of this evidence, however, concerns first-generation deterministic-regression models verified primarily against reanalysis (Section 2). Generative ensembles and high-resolution regional AI systems are largely absent from these comparisons.

Here we evaluate eleven forecast systems against the ECMWF IFS reference on European synoptic, solar-radiation and rain-gauge stations over ten months for 10 m wind, 2 m temperature, hourly shortwave accumulation and hourly precipitation, at lead times from one hour to two days. Skill is reported overall and in observed-value regimes defined as percentile exceedances relative to a 1991–2020 climatology, following the standard climate-extremes convention [Zhang et al., 2011, Perkins and Alexander, 2013] (Section 3.3); for precipitation we add categorical exceedance scores for the heavy tail. Section 3 describes the verification protocol.

*jua.ai

2 Related work

Evidence of AI deficits at extremes. Charlton-Perez et al. [2024] evaluated FourCastNet, FourCastNet-v2, Pangu-Weather and GraphCast on storm Ciarán against operational IFS analysis and ERA5. All four captured track and central pressure but underestimated peak winds by roughly 8 m s^{-1} . Forecasts initialised from ERA5 did not share this low wind bias, so smoothed training data is unlikely to be the cause; whether smoothed verification truth contributes remains open in that design. Olivetti and Messori [2024b] compared operational Pangu-Weather and GraphCast with IFS HRES against ERA5 regridded to 1.5° and found data-driven models competitive at extremes in most regions, but underestimated hot and wind extremes at longer leads. Zhang et al. [2026] reported physical HRES consistently ahead of GraphCast, Pangu-Weather and FuXi on record-breaking extremes defined from ERA5. Bonavita [2024] attributed the smoothing to a failure to reproduce sub-synoptic phenomena, with behaviour closer to post-processing than free-running simulation. Ben Bouallègue et al. [2024] is the main station-based exception: against SYNOP observations, Pangu-Weather was accurate for moderate extremes but oversmoothed the largest ones. Across these studies, every evaluated model is a deterministic-regression system from 2022–2024, and the most frequently tested system is also the oldest, NVIDIA’s FourCastNet [Pathak et al., 2022] (February 2022).

Training objectives and generative models. Subich et al. [2025] linked the deficit to the double penalty of grid-point MSE training, which reduced GraphCast’s effective resolution to roughly 1,250 km; an amplitude-preserving loss restored it to about 160 km and improved surface wind extremes. Molinaro et al. [2024] showed for turbulent flows that ensembles of MSE-trained networks regress to the mean flow, whereas score-based diffusion preserves statistics and spectra. Price et al. [2025] reported that GenCast, a generative global ensemble, outperforms the physical ENS on extremes against ERA5, and Pathak et al. [2026] demonstrated km-scale diffusion emulation of a regional convection-allowing model. A station-based comparison of these systems with the regression models above remains limited.

Station-based verification. WeatherReal [Jin et al., 2024] and MAUSAM [Gupta et al., 2025] provide in-situ benchmarks. MAUSAM finds forecast errors against observations 15 to 45% larger than against reanalysis, showing that reanalysis-centric verification can overstate skill. We verify extremes against stations using the StationBench verification stack [Schlueter and Wagner, 2025].

3 Methods

3.1 Models

Table 1 summarises the model set. We evaluate eleven systems against the ECMWF IFS high-resolution deterministic forecast as reference.

Physical baselines are ECMWF ENS, NOAA GFS, DWD ICON Global, and the regional DWD ICON-EU. Regression-family AI models are ECMWF AIFS [Lang et al., 2024], Microsoft Aurora [Bodnar et al., 2025], Jua EPT-2 Reasoning and EPT-2e [Molinaro et al., 2025, Jua AI, 2026], and the solar-specialised regional model Jua EPT-2.1 Helios [Molinaro, 2026]. Generative regional ensembles are Jua EPT-2 HRRR (5.5 km, 16 members, hourly updates; Siegenheim, 2025) and its successor Jua EPT-2.1 Europa (~ 7 km; Jua AI, 2026). Related generative methodology is described in Molinaro et al. [2026, 2024]. Figure 1 shows example 48 h fields from IFS HRES and EPT-2 HRRR. Precipitation is scored on the subset of systems that emit it: ECMWF IFS (reference), ECMWF ENS, NOAA GFS, DWD ICON Global and ICON-EU, and the Jua models EPT-2 Reasoning, EPT-2e, EPT-2 HRRR and EPT-2.1 Europa. ECMWF AIFS, Mi-

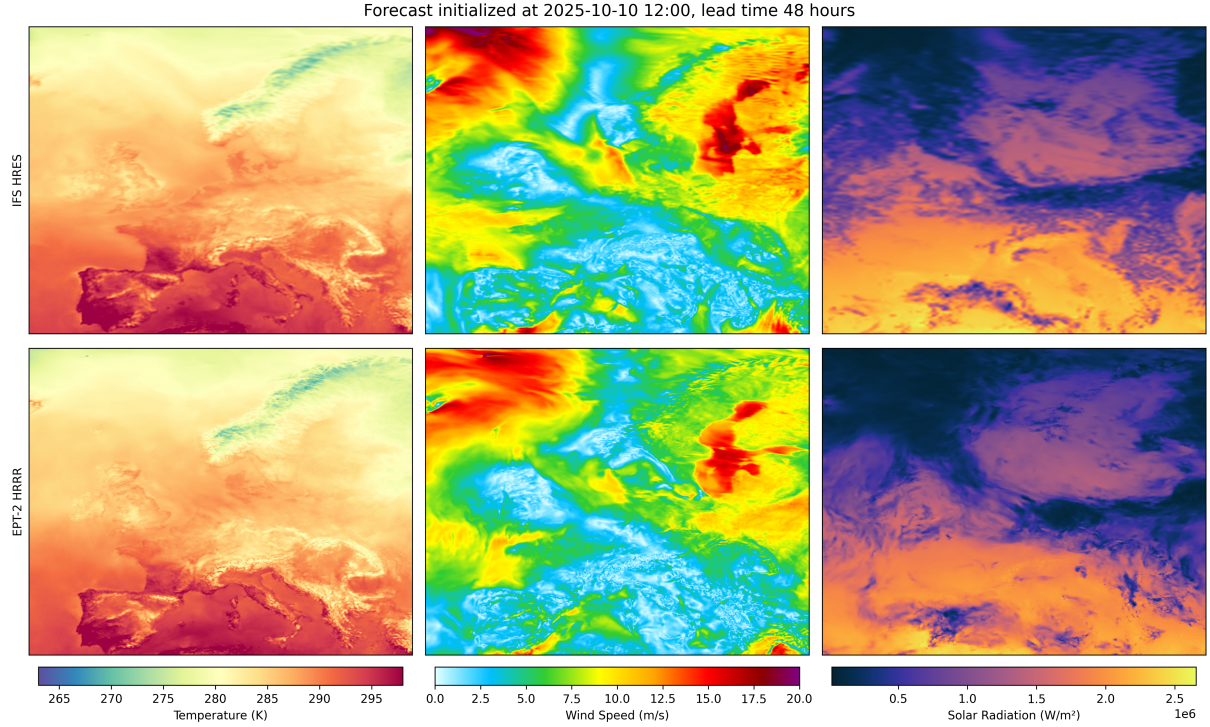


Figure 1: Forecast fields at 48 h lead (initialised 2025-10-10 12 UTC) from ECMWF IFS HRES (top) and Jua EPT-2 HRRR (bottom): 2 m temperature, 10 m wind speed, and shortwave accumulation. Illustrative fields from public model documentation [Siegenheim, 2025]; not used in the skill scores below and not regenerated by the repository figure scripts.

Table 1: Model set.

Model	Paradigm	Output	Domain
ECMWF IFS (reference)	physics	deterministic	global
ECMWF ENS	physics	ensemble	global
NOAA GFS	physics	deterministic	global
DWD ICON Global	physics	deterministic	global
DWD ICON-EU	physics	deterministic	regional
ECMWF AIFS	AI regression	deterministic	global
Microsoft Aurora	AI regression	deterministic	global
Jua EPT-2 Reasoning	AI regression	deterministic	global
Jua EPT-2e	AI regression	ensemble	global
Jua EPT-2 HRRR	AI generative	ensemble	regional
Jua EPT-2.1 Europa	AI generative	ensemble	regional
Jua EPT-2.1 Helios	AI regression (solar)	deterministic	regional

crosoft Aurora and the solar-only EPT-2.1 Helios produce no precipitation field and are omitted from that variable.

3.2 Station observations, variables, and evaluation

Verification uses quality-controlled in situ stations over Europe (Figure 2): a synoptic network for 10 m wind and 2 m temperature, and a separate solar-radiation network for hourly global horizontal irradiance (GHI). The latter is assembled from national and research networks, in-

cluding DWD, Météo-France, MeteoSwiss, KNMI, RMI, DMI, MET Norway (Frost), GeoSphere Austria, SMHI, FMI, BSRN, and ARPA Lombardia. Gridded forecasts from every model are interpolated to station locations by bilinear interpolation in latitude and longitude, as in Molinaro et al. [2026]. None of the Jua models are trained or fine-tuned on these synoptic or solar stations. The same station network is used for the rolling mean bias correction in Section 3.5. Skill scores are pooled over thirteen countries (DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, DK).

Hourly precipitation uses a separate rain-gauge panel (4,005 gauges) assembled from national networks: DWD (Germany), Météo-France, MET Norway, AEMET (Spain), GeoSphere Austria, the UK Met Office, KNMI, RMI, CHMI, DMI and MeteoSwiss. The synoptic feed does not carry a usable hourly precipitation accumulation. Gauge coverage spans eleven of the thirteen countries (all but IT and PL). Forecasts are interpolated to gauge locations as above, and each model is scored only where the gauge falls inside its forecast domain, so the regional models are compared on the gauges they actually cover. As with the other variables, none of the Jua models are trained or tuned on these gauges.

Two evaluation tracks follow model coverage:

- **Track A** covers 1 September 2025 to 30 June 2026 over the thirteen countries above. For wind and temperature we evaluate the nine models available throughout this window (ICON-EU is deferred to Track B) on a common 6 to 48 h lead grid at 6-hourly resolution, matching the cadence of AIFS and Aurora. For solar we use the same period on a common 1 to 48 h hourly lead grid, adding Jua EPT-2.1 Helios and omitting AIFS, Aurora and ENS, which have no usable solar output in this comparison. GB, ES and PL have no solar stations in this network; Italian solar coverage (mainly ARPA Lombardia) ends after February 2026, so March–June solar results omit IT. Hourly precipitation is evaluated on the same window and 1 to 48 h hourly grid, over the subset of systems that emit a precipitation field (Section 3.2; ECMWF ENS on its native 6-hourly cadence).
- **Track B** covers 1 March to 30 June 2026 and adds DWD ICON-EU for a regional comparison against the generative ensembles. Wind, temperature and solar (including EPT-2.1 Helios) share the 1 to 48 h hourly lead grid; precipitation is scored on the same grid for the regional emitters (EPT-2.1 Europa, EPT-2 HRRR, ICON-EU).

3.3 Observed-value regimes

We define regimes using a fixed 30-year climatology rather than percentiles of the evaluation window itself. Thresholds are derived from ERA5 [Hersbach et al., 2020] over 1991–2020, bilinearly interpolated from the 0.25° grid to each station location. For every station, calendar day and UTC hour we estimate P5, P25, P75 and P95 from a cyclic ± 15 -day calendar window, giving 930 samples (31 days $\times$ 30 years) behind each threshold and preserving both the seasonal cycle and the diurnal cycle. Each forecast–observation pair is then assigned to a regime by comparing its observed value with the thresholds for that station, day and hour. Solar thresholds use positive (daytime) irradiance only. Table 2 lists the six regimes; the physical labels are shorthand for the percentile bands. Defining extremes as percentile exceedances relative to a fixed climatological baseline is the standard convention of the climate-extremes literature: the ETCCDI indices and the WMO *Guidelines on Analysis of Extremes in a Changing Climate* [Zhang et al., 2011, Klein Tank et al., 2009] use exactly this construction for temperature and precipitation, and percentile exceedance relative to a local climatology is the accepted operational heatwave definition [Perkins and Alexander, 2013, Perkins, 2015]. Anchoring the thresholds to a 30-year (1991–2020) reference period follows the same WMO climate-normal convention, and we apply the framework to wind and irradiance as well. For wind and temperature the upper tail is the primary test. For solar, the low and high tails correspond to overcast and clear-sky conditions;

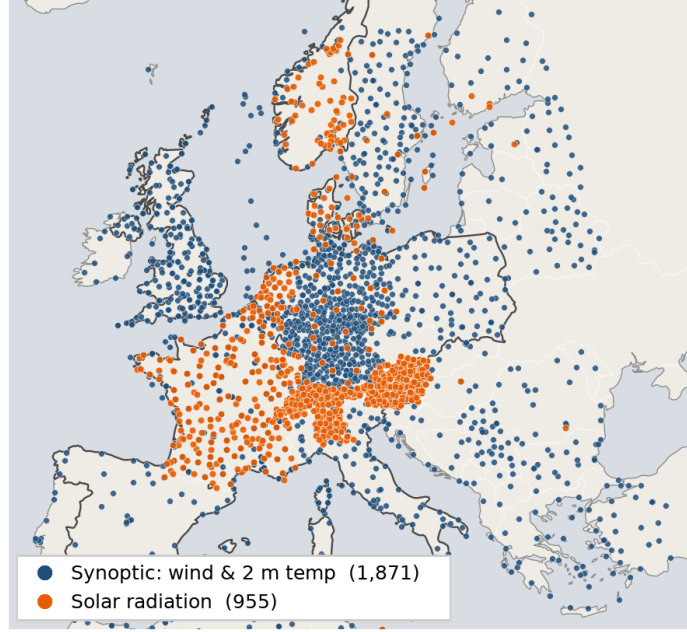


Figure 2: Station catalogues used for verification. Blue: synoptic network (10 m wind and 2 m temperature; $n = 1,871$). Orange: solar-radiation network ($n = 955$). Markers show the full catalogues; reported skill scores are pooled only over the thirteen countries listed in Section 3.2.

Table 2: The six observed-value regimes. Thresholds are P5/P25/P75/P95 of ERA5 1991–2020, interpolated to each station and estimated per station, calendar day and UTC hour, shared across models and leads. Solar: daytime accumulated GHI. Buckets follow the observation, not the forecast.

Regime	Percentiles	Wind	Temperature	Solar
All	–	all conditions	all conditions	all daytime
Very low	<P5	calm	severe cold	heavily overcast
Low	P5–P25	light air	cold	overcast
Typical	P25–P75	moderate	moderate	mixed sky
High	P75–P95	fresh to strong	warm	mostly sunny
Very high	>P95	gale force	heat	near clear sky

the lead-time figures focus on the overcast P5–P25 band, while the regime figures and tables report all six bands.

Because the thresholds are external to the scored period, regime occupancy is no longer fixed at 5/20/50/20/5% by construction; it instead measures how the evaluation period compared with 1991–2020. Temperature shows a clear warm displacement, with 13.4% of hours above the climatological P95 and only 3.3% below P5. Wind and irradiance instead show a symmetric widening to roughly 10% of hours in each tail because ERA5 is a gridded area average whose variance is smaller than that of point station observations. Regimes here are therefore defined relative to the gridded climatology, and the wind and solar tails are correspondingly broader than a station-native climatology would give.

Precipitation is zero-inflated, so the symmetric P5–P95 construction above degenerates (most percentiles collapse to zero) and is replaced by wet-hour climatological regimes (Table 3). Hours below 0.1 mm are grouped into a single *dry* regime; the remaining wet hours are split by P50, P75 and P95 of the ERA5 1991–2020 *wet-hour* distribution, estimated per station, calendar day and UTC hour on the same ± 15 -day window and interpolation as the other variables. Conditioning the percentiles on wet hours mirrors the ETCCDI wet-day precipitation indices (for example

Table 3: Wet-hour precipitation regimes. The dry regime collects hours below 0.1 mm; the wet regimes split hours at or above 0.1 mm by P50/P75/P95 of the ERA5 1991–2020 wet-hour distribution, per station, calendar day and UTC hour. Buckets follow the observation.

Regime	Definition	Interpretation
All	–	all hours
Dry	< 0.1 mm	dry hour
Light	wet, <P50	light rain
Moderate	P50–P75	moderate rain
High	P75–P95	heavy rain
Heavy	>P95	most intense hours

R95p), which likewise define precipitation extremes on the wet portion of the record rather than on the full zero-inflated distribution [Zhang et al., 2011, Klein Tank et al., 2009]. This isolates skill on the mass of the distribution (dry versus wet, and light versus heavy rain) rather than on percentiles that are almost all zero. As elsewhere, the boundaries are external to the scored period, so regime occupancy reflects how the evaluation period compared with 1991–2020.

3.4 Skill, lead times, and aggregation

Consider a model m in evaluation cell c (track, country, variable, lead, regime, debias setting). The skill is defined as

$$S_{m,c} = 1 - \frac{\text{MAE}_{m,c}}{\text{MAE}_{\text{IFS},c}}, \quad (1)$$

and is computed on matched samples only: the same country, regime, lead and initialisation intersection for every model in the cell. Forecasts are taken from all four daily cycles (00, 06, 12 and 18 UTC). For ensemble systems (ECMWF ENS, Jua EPT-2e, EPT-2 HRRR, EPT-2.1 Europa), MAE is computed on the ensemble mean; the analysis is therefore a point-forecast comparison and does not assess probabilistic calibration or spread. We also report mean forecast-minus-observation bias by regime. Results are reported at single leads (1, 6, 12, 24, 48 h where available) and as full-horizon aggregates over that comparison’s common lead grid only: 6 to 48 h at 6-hourly steps for Track A wind and temperature, and 1 to 48 h hourly for Track A solar and precipitation and for Track B. Models that lack a lead are omitted from that comparison; we do not interpolate onto another model’s grid. Uncertainty is reported as ± 1 leave-one-month-out jackknife standard error. For precipitation we additionally report categorical skill for the heavy (>P95) exceedance event, summarising the 2×2 contingency of forecast and observed exceedances pooled over gauges and leads to 48 h as probability of detection (POD), false-alarm ratio (FAR), critical success index (CSI) and frequency bias (forecast over observed event count). In line figures, colour identifies the model and line style identifies its class: solid for generative Jua systems, dash-dot for regression AI, and dashed for physical models (IFS remains solid black).

3.5 Debiasing

Station verification is sensitive to systematic, location-independent offsets that operational users often remove. For each model and hour of day, we estimate a mean bias from the previous four weeks of European station errors and subtract it from forecasts in the following week:

$$\tilde{f}_m(t) = f_m(t) - \bar{b}_m(h(t)), \quad \bar{b}_m(h(t)) = \frac{1}{|\mathcal{D}|} \sum_{\mathcal{D}} (f_m - y), \quad (2)$$

where y is the observation, $h(t)$ is the hour of day of the valid time, and $\mathcal{D}$ is the preceding four-week window at that hour. The correction does not depend on initialisation or lead. It is causal, identical for every model, and applied as a location shift to every ensemble member, preserving spread. Because regimes are defined on the observation (Section 3.3), debiasing does not re-label regimes. This correction removes only the mean bias; it does not force the bias within each regime to zero.

Precipitation is zero-bounded and right-skewed, so an additive shift is inappropriate (it can drive forecasts negative and it inflates MAE). We instead apply a *multiplicative* correction: for each model and valid week the forecast is scaled by the ratio of summed observed to summed forecast precipitation over the preceding four weeks (pooled over initialisation hour and lead, clipped to $[0.33, 3]$), which removes a systematic wet or dry amount bias while leaving zeros at zero. All precipitation results are reported after this correction.

4 Results

4.1 Skill by lead time

Figures 3–5 and 6 show MAE skill against ECMWF IFS by lead time for all four variables; Tables S8–S10 give the wind, temperature and solar values. Jua EPT-2.1 Europa holds $+8.1$ to $+8.7\%$ wind skill at every lead from 6 to 48 h, and Jua EPT-2 HRRR holds $+10.1$ to $+12.6\%$ for temperature. ECMWF AIFS is below the reference at every wind lead, from -6.6% at 6 h to -4.0% at 48 h. For solar (Figure 5), Jua EPT-2.1 Helios has the highest point estimate at every plotted lead, with the largest margin at the shortest leads. For precipitation, the Jua advantage generally emerges from 12 h onward, whereas GFS and ICON Global remain at or below IFS at every plotted lead.

4.2 Skill across the observed distribution

Aggregating over lead, Figures 7 and 9 show Track A MAE skill by observed-value regime for all four variables. Wind, temperature and solar use the six climatological bands in Figure 7 and Tables S1–S3 (1 to 12 h companions in Figure 8 and Tables S4–S6). Precipitation uses the dry mass plus four wet-hour bands in Figure 9 and Table 5, with a 1 to 12 h companion in Figure 10 and Table S7. Thus every variable is evaluated across its full observed distribution, including the very-low or dry regime.

For wind, EPT-2.1 Europa peaks in the typical regime ($+11.3 \pm 0.7\%$), while at gale force DWD ICON Global has the highest point estimate ($+9.5 \pm 1.2\%$), effectively tied with EPT-2.1 Europa ($+9.0 \pm 1.2\%$). NOAA GFS is barely positive there ($+0.6 \pm 0.7\%$) and poor under all conditions (-11.7% wind, -23.9% temperature). For temperature, EPT-2 HRRR is marginally ahead overall ($+12.1 \pm 0.4\%$ versus EPT-2.1 Europa $+11.5 \pm 0.7\%$) and also leads in the heat regime ($+19.6 \pm 2.2\%$ versus EPT-2.1 Europa $+15.0 \pm 1.7\%$), though that gap is within the combined jackknife uncertainty.

Jua EPT-2 Reasoning is positive under all conditions and through the middle of the distribution but only marginally so in the wind tails ($-0.7 \pm 0.3\%$ at gale force) and clearly negative at cold extremes ($-6.9 \pm 1.3\%$); Microsoft Aurora and Jua EPT-2e stay within about 2% of IFS on the all-conditions Track A pool but lose more than 20% in the cold tail. ECMWF AIFS is negative in every reported wind regime, reaching $-10.6 \pm 0.6\%$ at gale force, and loses $4.9 \pm 2.0\%$ at heat extremes despite near-neutral overall temperature skill. Per-country heat skill (Table 4) shows NOAA GFS negative in every country with data.

On the same September–June window, solar on the hourly 1 to 48 h pool (EPT-2.1 Helios included; AIFS, Aurora and ENS omitted) shows no uniform smoothing failure. EPT-2.1 Helios has the highest all-conditions point estimate at $+10.2 \pm 1.7\%$, ahead of EPT-2.1 Europa ($+7.1 \pm 1.0\%$) and EPT-2 Reasoning ($+6.3 \pm 0.5\%$). Its margin is concentrated in the tails: $+24.8 \pm$

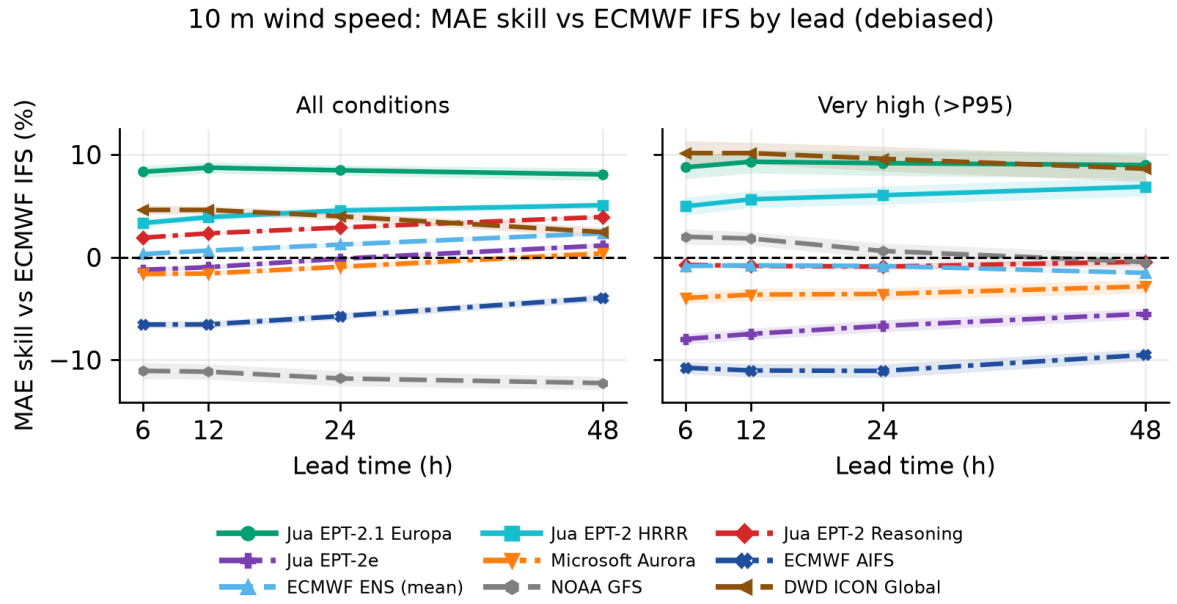


Figure 3: MAE skill against ECMWF IFS by lead time (6, 12, 24, 48 h), 10 m wind, debiased. Left: all conditions; right: very high (>P95, gale-force). Bands show ± 1 jackknife SE.

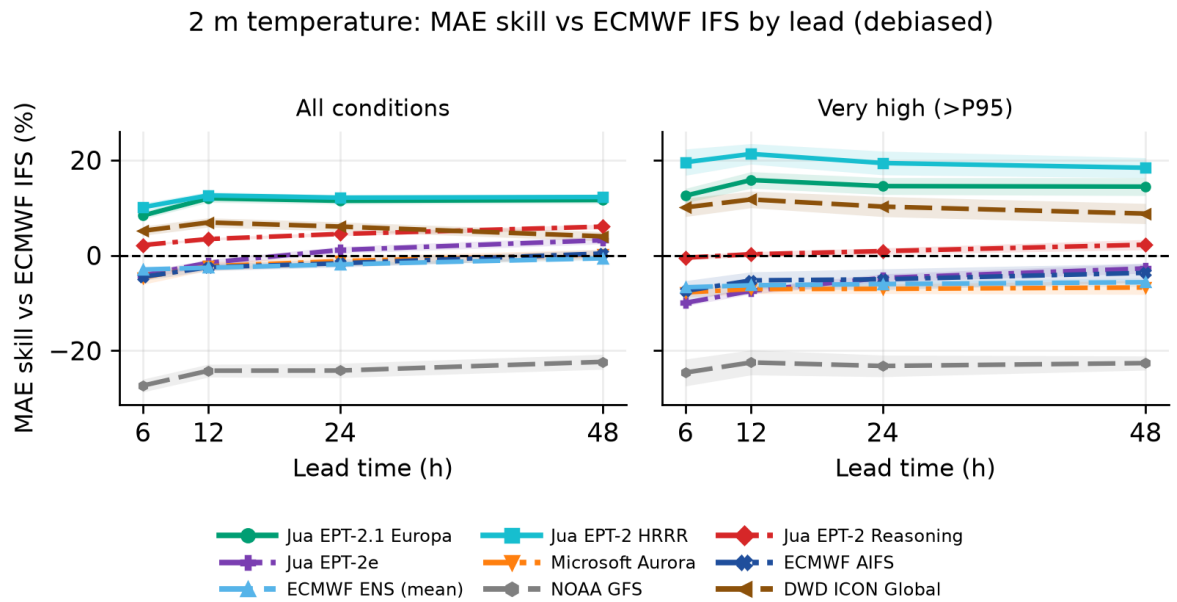


Figure 4: As Figure 3, for 2 m temperature (right: very high / heat, >P95).

1 h shortwave accumulation: MAE skill vs ECMWF IFS by lead (debiased)

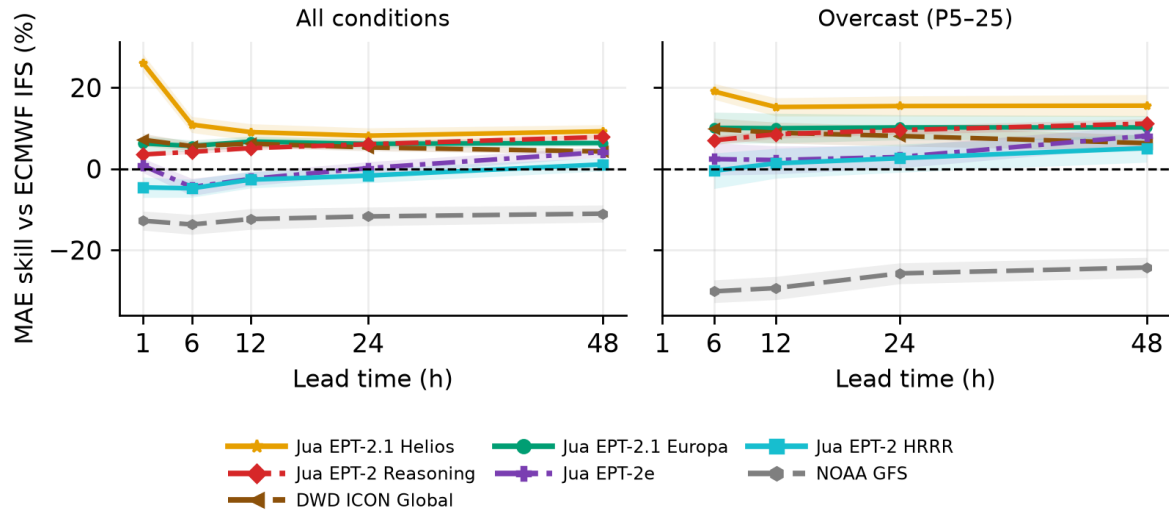


Figure 5: MAE skill against ECMWF IFS by lead time for 1 h shortwave accumulation on the Track A September 2025 to June 2026 window, debiased. Left: all conditions; right: the overcast tail (P5–P25), where EPT-2.1 Helios’s tail specialisation gives it a large positive margin at every lead. Leads are 1, 6, 12, 24 and 48 h. AIFS, Aurora and ENS are omitted (no usable solar output in this comparison).

1 h precipitation: MAE skill vs ECMWF IFS by lead (debiased)

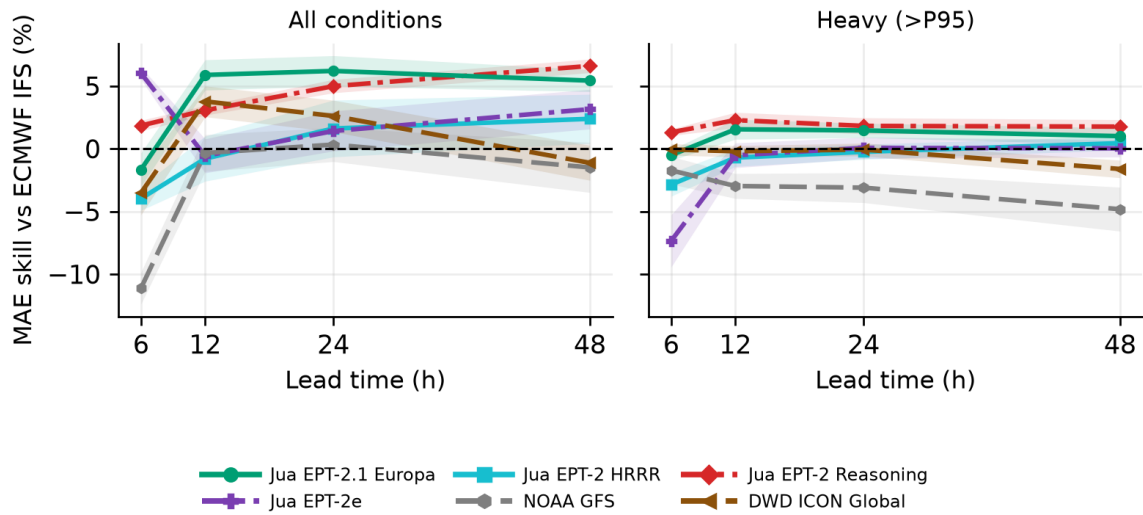


Figure 6: MAE skill against ECMWF IFS by lead time (6, 12, 24, 48 h) for 1 h precipitation, Track A, debiased. Left: all conditions; right: heavy (>P95). Bands show ± 1 jackknife SE.

Table 4: Track A per-country MAE skill (%) at heat extremes ($>P95$ temperature), 6 to 48 h, debiased. Point estimates. NOAA GFS is negative in every listed country.

Model	DE	FR	GB	ES	IT	PL	NL	BE	CH	NO	AT	CZ	DK
Jua EPT-2.1 Europa	+6.6	+16.1	-1.2	+6.4	+17.2	+6.2	-3.4	-3.0	+31.1	+5.5	+28.2	+16.6	-4.4
Jua EPT-2 HRRR	+10.5	+21.5	+4.0	+10.5	+27.6	+5.6	+1.8	-0.6	+29.4	+14.2	+16.6	+17.7	+12.2
Jua EPT-2 Reasoning	+6.6	+2.6	+4.3	-2.6	+0.1	+3.9	+3.8	-1.0	-1.3	-5.2	-3.1	+7.5	+2.7
Jua EPT-2e	+6.1	-2.2	-0.3	-15.1	-10.4	-2.4	+0.6	-2.8	-2.5	-12.2	-16.3	-1.3	+1.7
Microsoft Aurora	+2.1	-4.5	-7.5	-18.7	-7.8	-0.9	+0.7	-11.8	-3.3	-18.0	-15.4	-7.0	+1.8
ECMWF AIFS	+0.8	-1.6	-10.2	-12.7	-14.6	+1.9	-0.4	+6.3	+1.9	-3.4	-25.0	-3.1	+10.3
ECMWF ENS (mean)	-2.1	-4.6	-5.1	-12.1	-4.0	-8.6	+3.4	-2.9	-10.7	-9.8	-10.6	-1.5	-9.2
NOAA GFS	-18.4	-23.6	-37.0	-25.1	-13.2	-24.3	-21.9	-12.2	-10.2	-43.1	-25.3	-31.4	-24.7
DWD ICON Global	+6.9	+8.8	-4.5	+3.6	+12.8	+4.8	-2.7	+10.0	+22.7	+18.1	+16.8	+15.5	-6.5
DWD ICON-EU [†]	+13.1	+18.4	+1.3	+7.4	+20.1	+9.1	-2.6	+14.1	+34.6	+6.6	+28.1	+17.8	-0.7

5.4% in the clear-sky tail ($>P95$) and $+16.4 \pm 3.4\%$ at overcast ($P5$ – $P25$). In the typical irradiance regime ($P25$ – $P75$), where cloud-driven variability is largest, EPT-2.1 Helios is instead the weakest of the AI systems ($-8.2 \pm 1.1\%$) and EPT-2 Reasoning leads ($+14.0 \pm 0.7\%$); the specialised model buys tail accuracy at the cost of the bulk of the distribution. EPT-2 HRRR is near-neutral overall on solar ($+0.6\%$) and mildly negative in the clear-sky tail ($-4.1 \pm 4.9\%$). Restricting to 1 to 12 h (Figure 8) widens EPT-2.1 Helios’s solar lead to $+13.5 \pm 1.9\%$, leaves EPT-2.1 Europa first on wind while swapping EPT-2 HRRR and ICON Global in second and third place, and lowers all-conditions temperature skill for both leaders (EPT-2 HRRR $+9.5 \pm 0.6\%$, EPT-2.1 Europa $+8.3 \pm 0.8\%$).

Precipitation completes the Track A regime comparison using the wet-hour bands of Section 3.3 (Figure 9, Table 5; lead curves in Figure 6). Its skill has a different shape from wind, temperature and solar: the gains sit in the middle of the wet distribution rather than at the extreme tail. Through the bulk of the wet distribution the Jua models beat IFS by a clear margin: in the moderate $P50$ – $P75$ band Jua EPT-2 HRRR reaches $+15.2 \pm 2.2\%$, EPT-2e $+14.8 \pm 1.7\%$ and EPT-2 Reasoning $+14.0 \pm 0.9\%$, with $+9$ to $+11\%$ still in the high ($P75$ – $P95$) band. EPT-2 Reasoning leads all conditions ($+5.1 \pm 0.4\%$); the physical global models are negative through the wet bands (NOAA GFS $-4.3 \pm 1.0\%$ overall, DWD ICON Global $-14.4 \pm 3.0\%$ at $P50$ – $P75$). On the 1 to 12 h pool (Figure 10), the three Jua systems retain $+8.2$ to $+11.2\%$ skill at $P50$ – $P75$, while their $>P95$ results already diverge (EPT-2 Reasoning $+2.2 \pm 0.5\%$, EPT-2 HRRR $-4.5 \pm 0.8\%$, EPT-2e $-6.0 \pm 1.3\%$).

At the heavy $>P95$ tail the broad Jua advantage narrows and becomes model-specific. EPT-2 Reasoning remains ahead at $+1.7 \pm 0.5\%$; EPT-2.1 Europa ($+0.2 \pm 0.6\%$) is near IFS; EPT-2 HRRR ($-1.5 \pm 0.5\%$) and EPT-2e ($-1.9 \pm 0.9\%$) trail it; and GFS is lower still ($-3.7 \pm 1.2\%$). These effects are much smaller than the 9–15% gains from $P50$ to $P95$, but several remain clearly separated from zero relative to their jackknife SE. Separately, the bias diagnostics show that all models, IFS included, under-forecast the most intense hours by close to 2 mm. This is the same centre-seeking conditional bias seen for wind and temperature (Section 4.3), not evidence that the models have identical relative MAE. Jua EPT-2.1 Europa is a model-specific exception in the other direction: it scores strongly on dry mass ($+21.0 \pm 2.7\%$) but is negative through the light and moderate wet bands (-11.1 to -10.5%), the signature of a systematic wet-amount over-forecast that the multiplicative debiasing only partly removes. Per-country precipitation profiles are shown in Figure S4.

Categorical detection of the heavy ($>P95$) event adds a complementary view (Figure 11, Table S19): EPT-2.1 Europa, ICON Global and EPT-2 Reasoning match or slightly exceed the IFS critical success index (0.19, 0.19, 0.18 versus 0.17), but every model under-detects heavy hours (frequency bias below one throughout, from 0.82 for ICON Global down to 0.29 for the ECMWF ENS mean), confirming that the shortfall at the wet extreme is shared rather than specific to the AI systems.

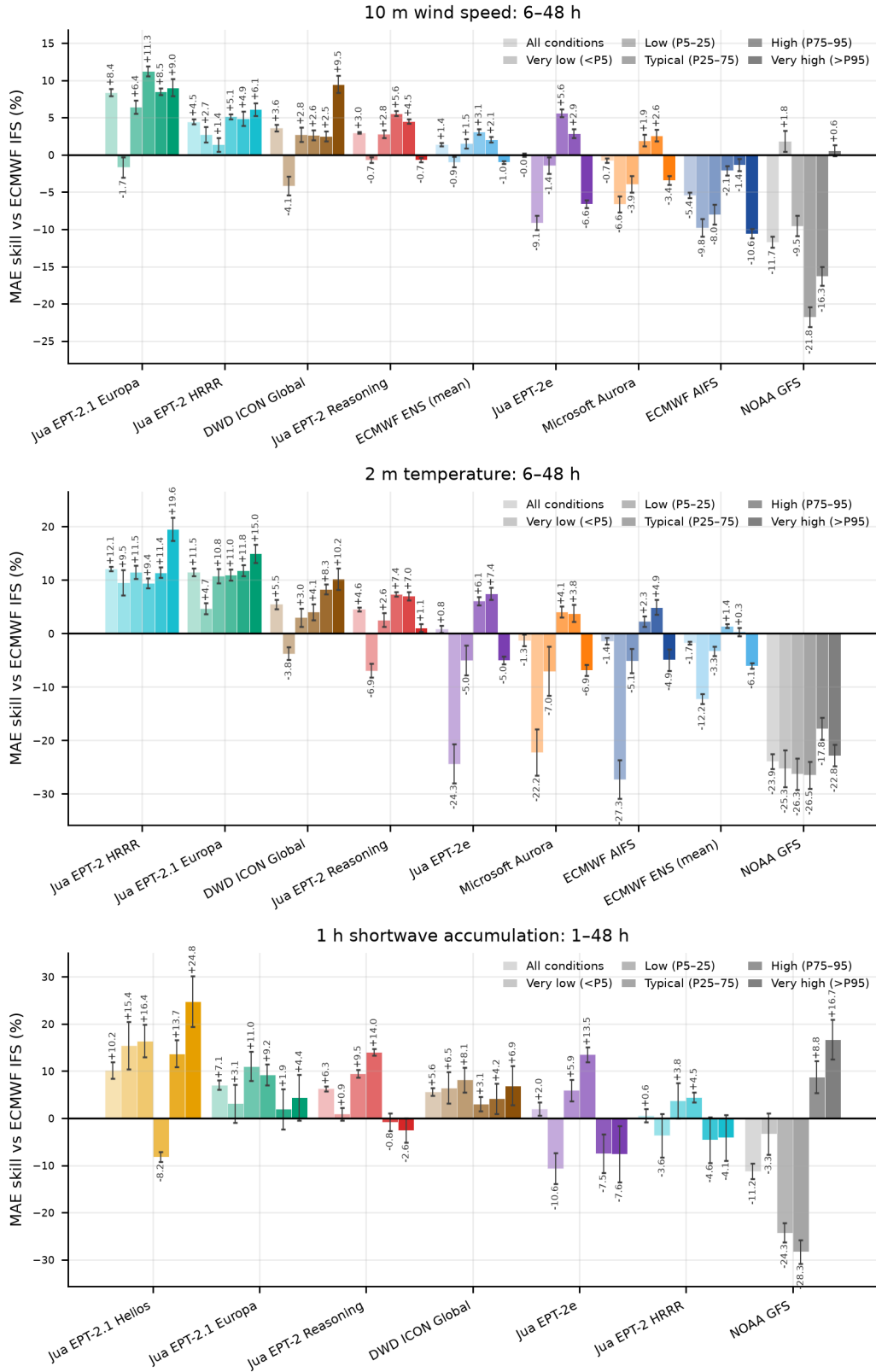


Figure 7: Track A MAE skill against ECMWF IFS with ± 1 jackknife SE, by observed-value regime, debiased. Wind and temperature: 6 to 48 h (6-hourly grid). Solar: 1 to 48 h hourly (EPT-2.1 Helios added; AIFS/Aurora/ENS omitted). September 2025 to June 2026. Regimes are the ERA5 1991–2020 climatological percentiles of Section 3.3. Models sorted by all-conditions skill within each panel.

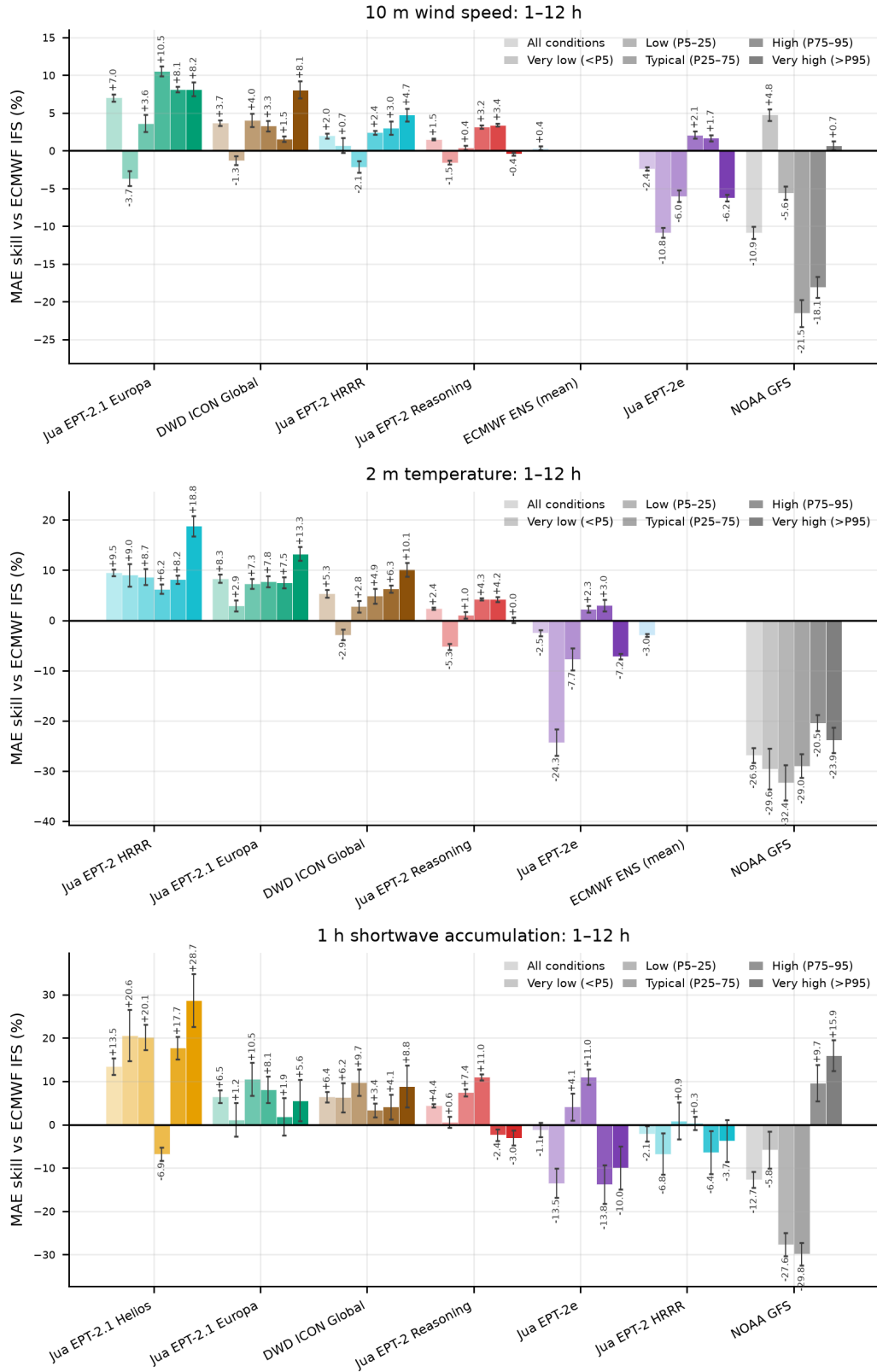


Figure 8: Track A MAE skill against ECMWF IFS with ± 1 jackknife SE by regime on the 1 to 12 h hourly pool, debiased. Wind and temperature use the hourly subset of the Track A models (AIFS and Aurora omitted; 6-hourly cadence only). Solar includes EPT-2.1 Helios. All six regimes are shown.

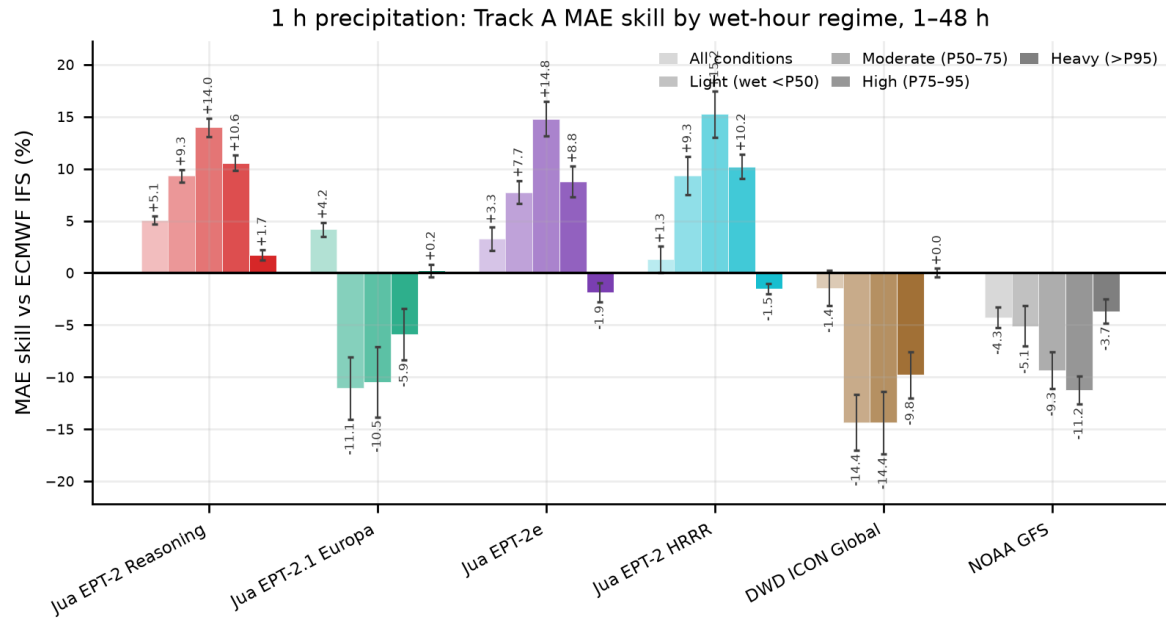


Figure 9: Track A MAE skill against ECMWF IFS with ± 1 jackknife SE for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased, September 2025 to June 2026. Bars are all conditions and the four wet bands (light <P50, moderate P50–75, high P75–95, heavy >P95); the dry regime is in Table 5. Models sorted by all-conditions skill.

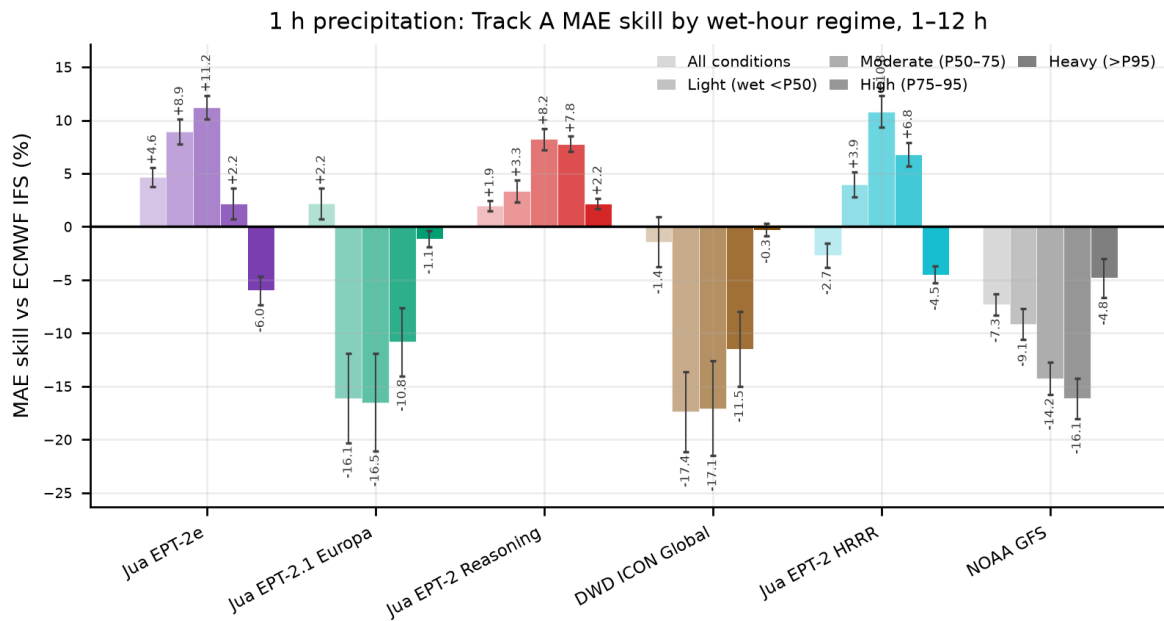


Figure 10: As Figure 9, restricted to the 1 to 12h hourly pool.

Table 5: Track A MAE skill (%) against ECMWF IFS for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased, with ± 1 jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+5.1 (± 0.4)	+2.2 (± 1.3)	+9.3 (± 0.6)	+14.0 (± 0.9)	+10.6 (± 0.8)	+1.7 (± 0.5)
Jua EPT-2 HRRR	+1.3 (± 1.3)	-6.1 (± 3.3)	+9.3 (± 1.8)	+15.2 (± 2.2)	+10.2 (± 1.2)	-1.5 (± 0.5)
Jua EPT-2e	+3.3 (± 1.1)	+1.5 (± 4.8)	+7.7 (± 1.1)	+14.8 (± 1.7)	+8.8 (± 1.5)	-1.9 (± 0.9)
Jua EPT-2.1 Europa	+4.2 (± 0.7)	+21.0 (± 2.7)	-11.1 (± 3.0)	-10.5 (± 3.4)	-5.9 (± 2.5)	+0.2 (± 0.6)
NOAA GFS	-4.3 (± 1.0)	-0.2 (± 1.8)	-5.1 (± 2.0)	-9.3 (± 1.8)	-11.2 (± 1.4)	-3.7 (± 1.2)
DWD ICON Global	-1.4 (± 1.7)	+8.3 (± 2.7)	-14.4 (± 2.7)	-14.4 (± 3.0)	-9.8 (± 2.2)	+0.0 (± 0.4)

1 h precipitation: heavy-precip exceedance, Track A 1–48 h (debiased)

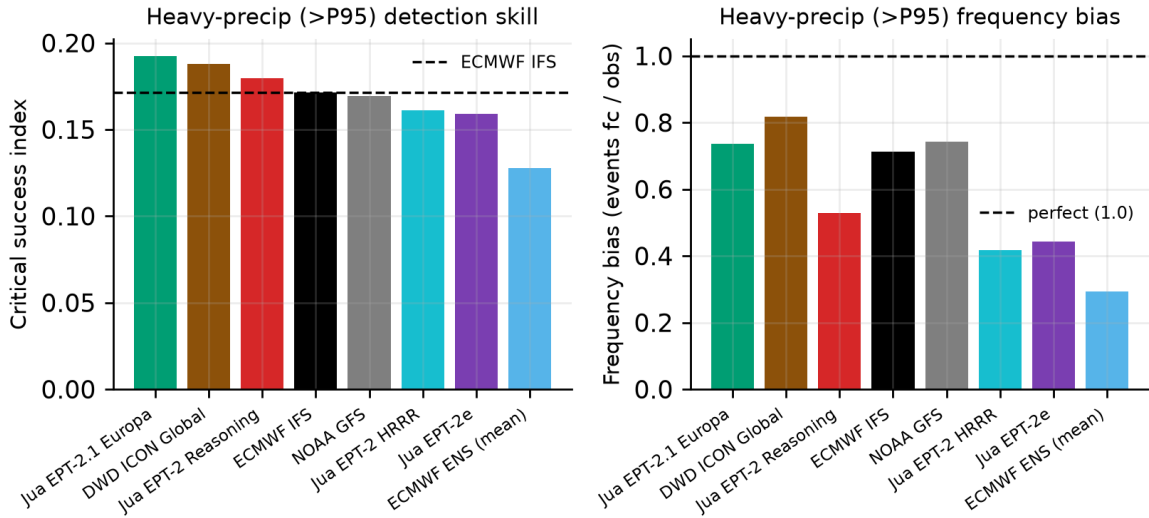


Figure 11: Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased. Left: critical success index (dashed line: ECMWF IFS). Right: frequency bias (forecast over observed event count; dashed line: perfect = 1). Every model under-forecasts the frequency of heavy hours.

4.3 Shared conditional bias

Figure 12 shows conditional bias by regime at 48 h after debiasing. The grey band is the min–max envelope across all models; ECMWF IFS (black), EPT-2.1 Europa (green) and AIFS (blue) are highlighted, with EPT-2.1 Helios (amber) in the solar panel and EPT-2 Reasoning (red) in the precipitation panel. All evaluated systems overforecast low observed values and underforecast high observed values. Wind bias is positive in the calm regime ($+0.8$ to $+1.2$ m s^{-1}) and negative in the gale regime (-2.2 to -2.9 m s^{-1}). Temperature has the same centre-seeking shape: warm bias in severe cold ($+0.7$ to $+1.2$ $^{\circ}\text{C}$) and cold bias in the heat regime (-0.3 to -0.7 $^{\circ}\text{C}$). Precipitation ranges from a $+0.09$ to $+0.12$ mm wet bias in light rain to a -2.29 to -2.51 mm dry bias at $>P95$.

4.4 The regional comparison

Track B compares the generative regional models with DWD ICON-EU on the March–June window and the hourly 1 to 48 h horizon (Figure 15; 1 to 12 h in Figure 16; leads in Figure 17; precipitation in Figures 13 and 14; Tables S11–S13 and S14–S16; precipitation values in Tables S18 and S17).

On the 1 to 48 h pool, EPT-2.1 Europa leads wind overall ($+7.2 \pm 0.4\%$ versus ICON-

Conditional bias by observed-value regime at 48 h (debiased)

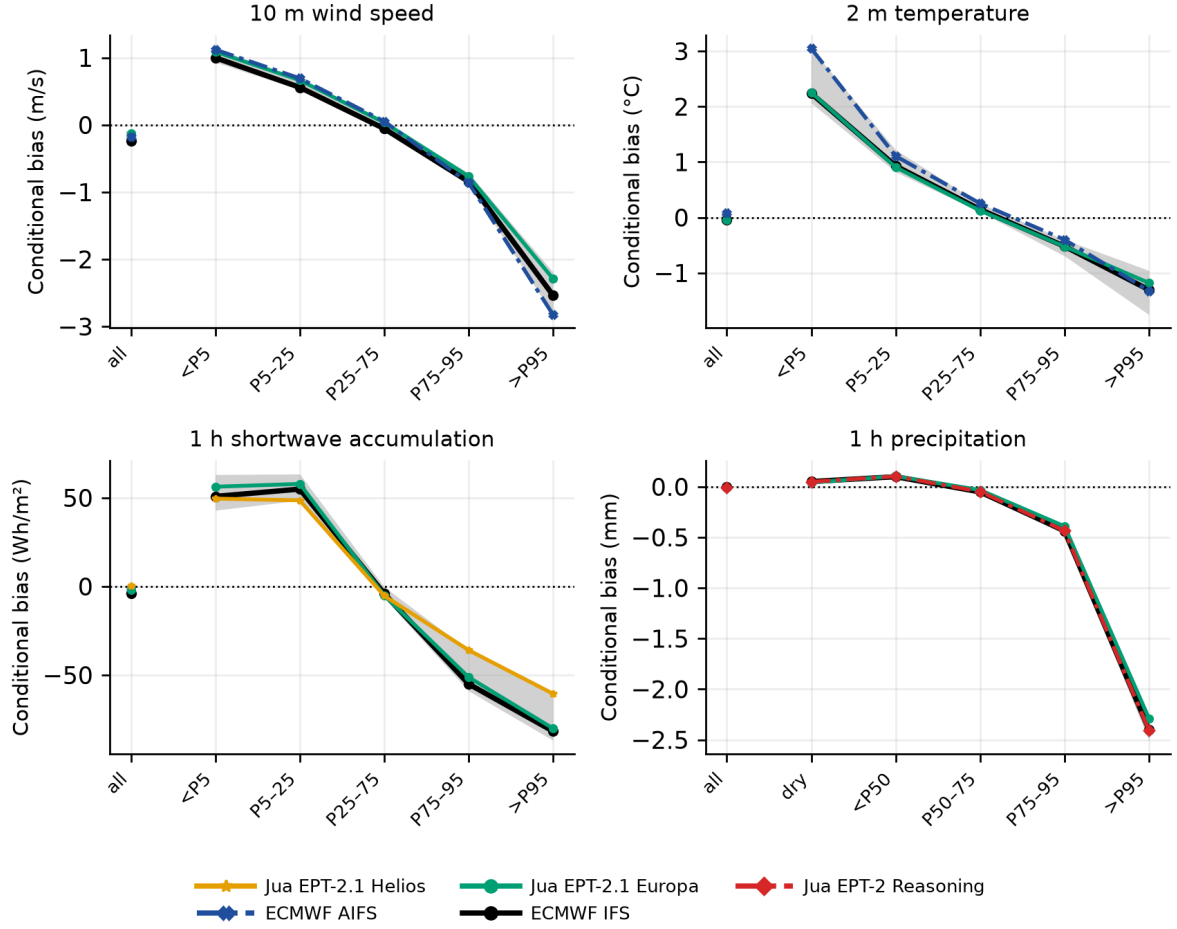


Figure 12: Conditional bias by observed-value regime at 48 h after debiasing. Grey band: min–max across all models. Highlighted: ECMWF IFS (black), Jua EPT-2.1 Europa (green), ECMWF AIFS (blue); solar panel adds Jua EPT-2.1 Helios (amber), and precipitation adds Jua EPT-2 Reasoning (red). The evaluated models overforecast low observed values and underforecast high observed values.

EU $+5.3 \pm 0.5\%$); at gale wind EPT-2.1 Europa and ICON-EU are level ($+10.5 \pm 1.4\%$ and $+11.5 \pm 1.6\%$), while EPT-2 HRRR is lower ($+4.1 \pm 1.5\%$). For temperature, EPT-2.1 Europa has the highest overall point estimate ($+13.2 \pm 0.6\%$, versus ICON-EU $+12.4 \pm 0.8\%$ and EPT-2 HRRR $+12.0 \pm 0.7\%$); all three overlap in the heat tail at about 17%. For solar, EPT-2.1 Helios has the highest overall point estimate ($+12.6 \pm 2.5\%$, versus EPT-2.1 Europa $+7.8 \pm 2.0\%$ and ICON-EU $+7.2 \pm 1.4\%$) and leads both the overcast and clear-sky tails; EPT-2 HRRR is near-neutral overall and negative in the clear-sky tail ($-6.4 \pm 7.7\%$). At 1 to 12 h, ICON-EU improves relative to the generative systems on temperature, EPT-2.1 Helios’s solar margin widens, and wind remains EPT-2.1 Europa-led. Lead curves (Figure 17) preserve the all-conditions wind ordering across the horizon, while gale and heat differences among the leading systems remain small relative to uncertainty. ICON-EU therefore matches the leading generative regionals in the wind and heat tails, while EPT-2.1 Europa retains the all-conditions wind lead. This comparison does not isolate resolution from other model differences.

Precipitation completes the same matched regional comparison (Figure 13, Table S18). EPT-2 HRRR is the only regional system with positive skill from light through high rain, reaching $+15.6 \pm 6.1\%$ at P50–P75. ICON-EU is positive overall ($+3.5 \pm 0.5\%$) but negative in each of

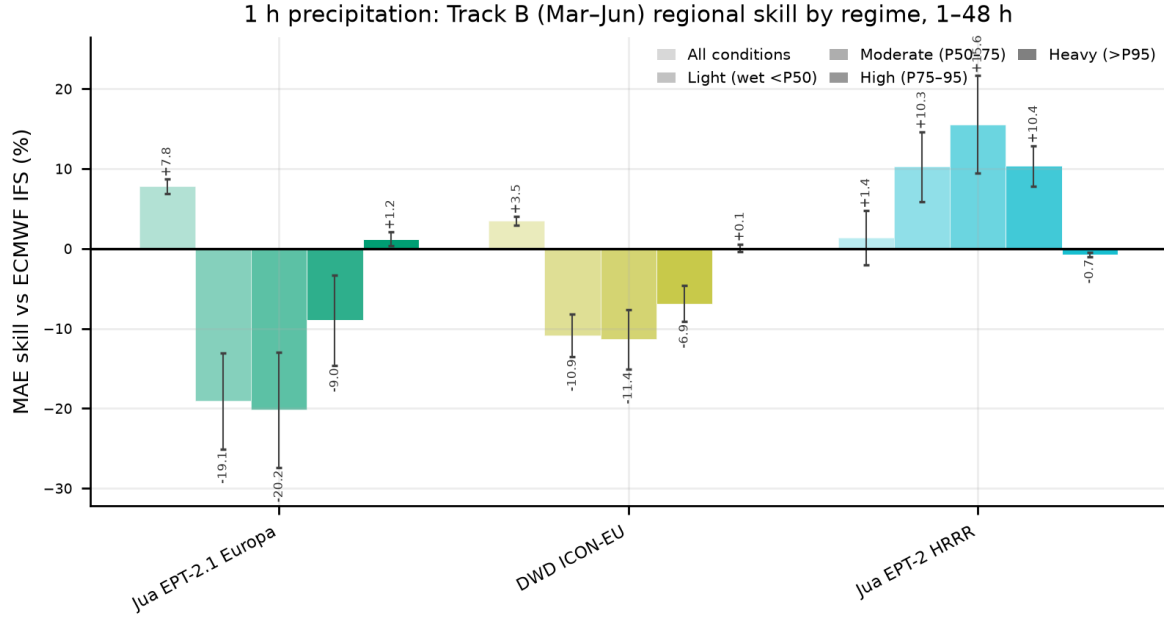


Figure 13: Track B (March–June 2026) MAE skill against ECMWF IFS for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased. EPT-2 HRRR is positive from light through high rain; heavy-tail effect sizes range from -0.7 to $+1.2\%$. Bars as in Figure 9.

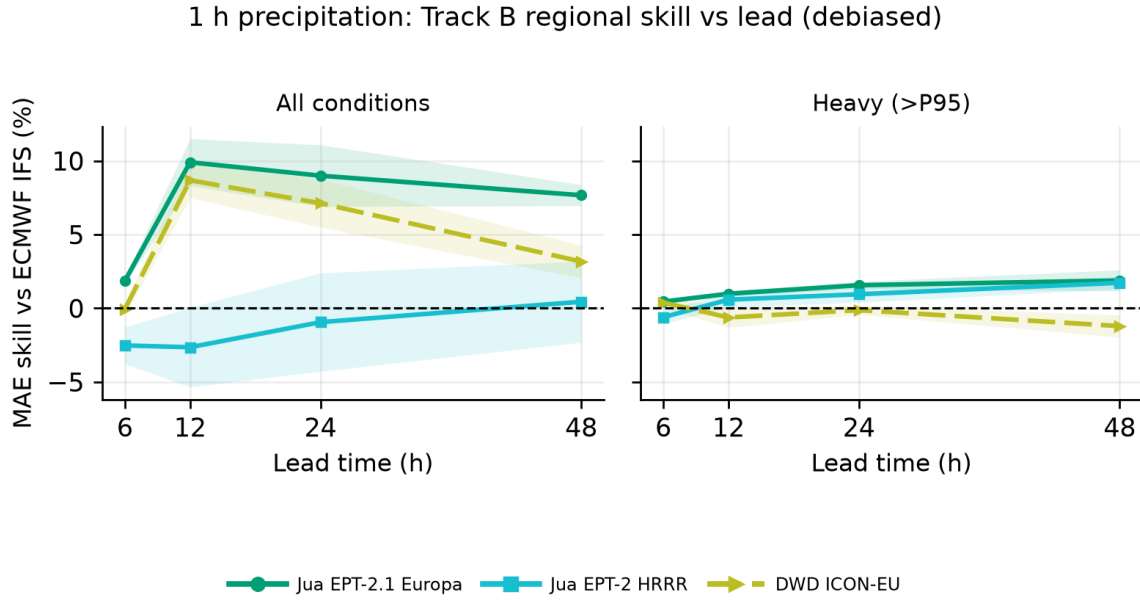


Figure 14: Track B precipitation MAE skill against ECMWF IFS by lead, debiased. Left: all conditions; right: heavy (>P95). Bands show ± 1 jackknife SE.

those wet bands, while EPT-2.1 Europa repeats its Track A signature: strong on dry mass and negative from light through high rain. At the heavy >P95 tail, effect sizes are small (EPT-2.1 Europa $+1.2 \pm 0.8\%$, EPT-2 HRRR $-0.7 \pm 0.3\%$, ICON-EU $+0.1 \pm 0.5\%$), although EPT-2 HRRR remains clearly below zero relative to its SE.

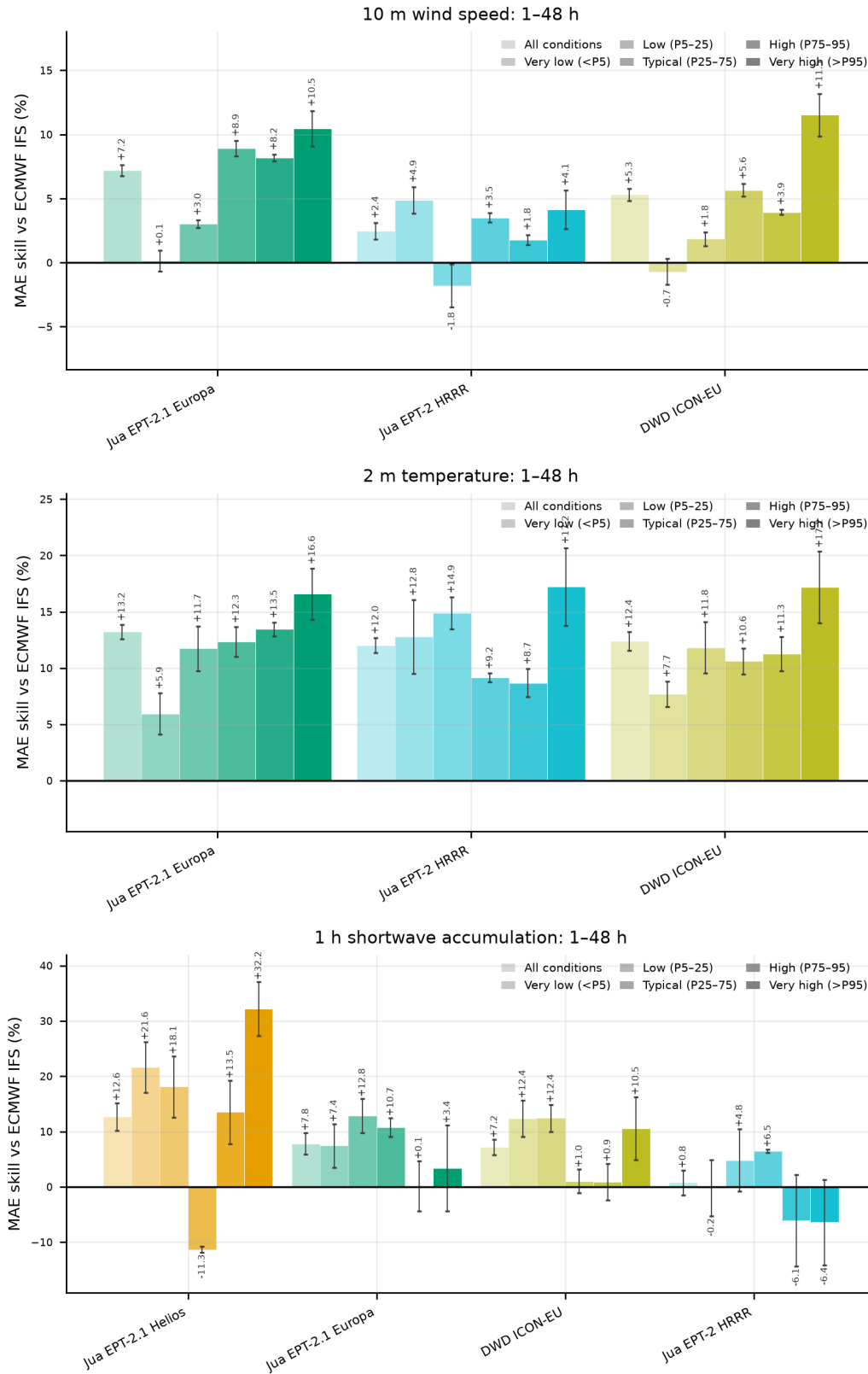


Figure 15: Track B MAE skill against ECMWF IFS with ± 1 jackknife SE, by observed-value regime, debiased, March to June 2026, 1 to 48 h. Top and middle: 10 m wind and 2 m temperature (EPT-2.1 Europa, EPT-2 HRRR, ICON-EU). Bottom: solar for that set plus EPT-2.1 Helios. Models sorted by all-conditions skill within each panel.

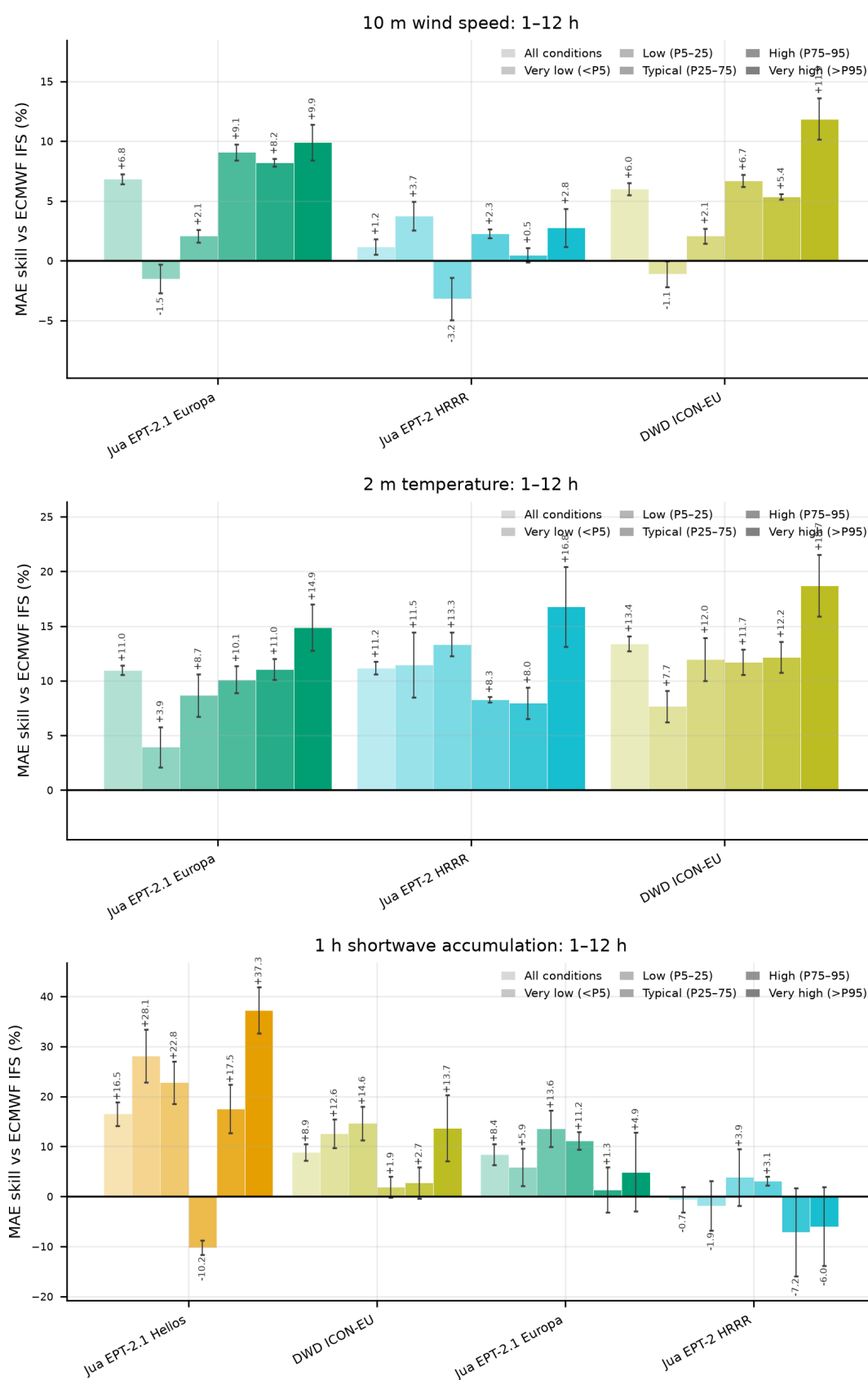


Figure 16: As Figure 15, restricted to the 1 to 12h hourly pool.

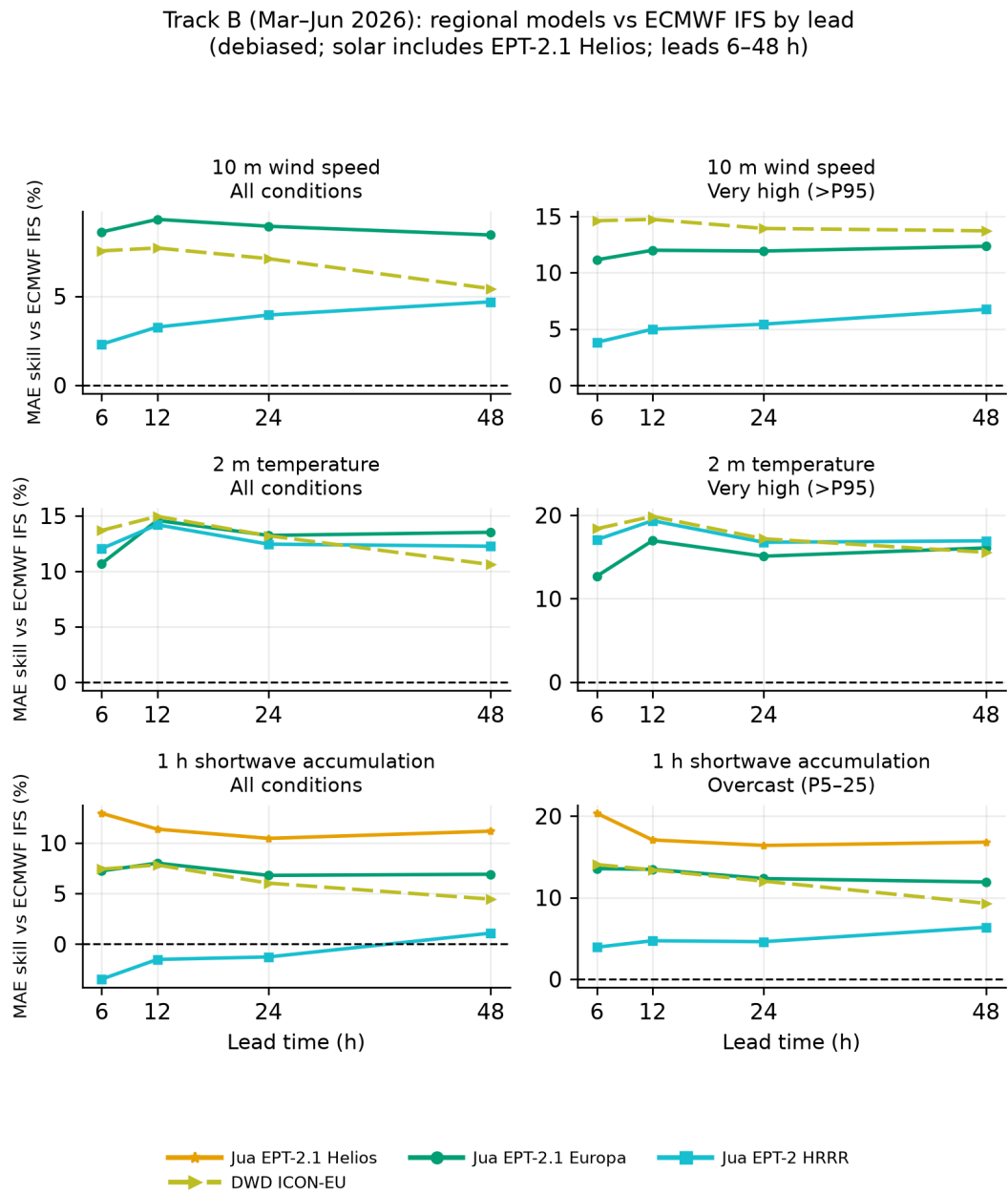


Figure 17: Track B, March to June 2026: regional models against ECMWF IFS by lead, debiased. Top: 10 m wind (all conditions and >P95). Middle: 2 m temperature (all conditions and >P95). Bottom: 1 h shortwave including EPT-2.1 Helios (all conditions and overcast P5-25). Plotted leads are 6, 12, 24 and 48 h.

4.5 Per-country regime profiles

Figures S1–S4 show MAE skill by regime in each country on the March–June 2026 window, pooled over hourly leads 1 to 48 h. The model set mixes the Track B regionals (EPT-2.1 Europa, EPT-2 HRRR, ICON-EU, EPT-2.1 Helios) with selected global baselines (EPT-2 Reasoning, ICON Global, GFS), so these panels are neither the matched Track A nor the matched Track B comparison. Regimes use the same ERA5 1991–2020 climatological percentiles of Section 3.3 as the rest of the paper, so their bucket boundaries are directly comparable with the preceding results. Country panels largely reproduce the Europe-pooled ordering: EPT-2.1 Europa and ICON-EU remain among the strongest wind and temperature systems in most panels. EPT-2.1 Helios leads solar in most countries. GFS heat and all-conditions temperature deficits appear widely rather than in a single country. Countries without sufficient matched samples for a variable are omitted (GB, IT and PL on solar, which lack solar station observations; IT and PL on precipitation, which lack gauges).

5 Discussion

Prior AI-extremes work focused on first-generation regression models verified against smoothed reanalysis and already reported competitive performance at extremes in many regions, with a remaining deficit in event magnitude [Olivetti and Messori, 2024b]. Against independent station observations and climatological observed-value regimes, several systems here have lower MAE than IFS in the heat and gale tails. EPT-2.1 Helios improves on IFS in both solar tails but is the weakest AI system in the typical irradiance regime, so specialisation does not transfer uniformly across the distribution. The Jua precipitation models improve most clearly through the middle of the wet distribution, not at its heavy tail. Across all four variables, relative tail skill varies among both AI and physical models.

The deficits are model-specific rather than tied to the AI/physical split or to the training objective. ECMWF AIFS carries clear heat- and wind-tail deficits, yet the also regression-trained Jua EPT-2 Reasoning carries much smaller ones, and the largest heat-tail deficit of all belongs to a physical model, NOAA GFS, which is only marginally positive at gale wind. Among the regional systems, ICON-EU matches EPT-2.1 Europa in the wind and heat tails and is competitive on solar; for precipitation, EPT-2 HRRR alone remains positive from light through high rain. Because the models also differ in architecture, training data and initialisation, this comparison does not isolate the effect of resolution [Bonavita, 2024, Subich et al., 2025].

The most consistent cross-variable result is the shared conditional bias. After debiasing, every system, AI and physical, overforecasts low observed values and underforecasts high ones (Section 4.3). This common bias does not erase model-specific relative skill: at heavy precipitation EPT-2 Reasoning is modestly ahead of IFS, EPT-2.1 Europa is near it, and EPT-2 HRRR, EPT-2e and GFS trail it. The shared centre-seeking error is therefore a property of the full model class, while its MAE cost still differs by system.

6 Limitations

Thresholds come from a 30-year reanalysis climatology rather than from station records, so “extreme” means rare relative to the ERA5 1991–2020 distribution at that station, day and hour, not necessarily an impact-defined heatwave or gale. Because ERA5 is a gridded area average, its variance is smaller than that of point observations; for 10 m wind and surface irradiance this widens both tails to roughly 10% occupancy each rather than the nominal 5%, so those tails are less selective than the temperature tails.

Precipitation is scored on a separate rain-gauge panel covering eleven countries (Italy and Poland are absent), and against ERA5 wet-hour thresholds that, like the other variables, describe

a gridded area average rather than a point gauge; hourly gauge accumulations are also noisier than the synoptic wind and temperature reports. The heavy-precipitation results should be read as a relative comparison against IFS on this panel, not as an absolute measure of convective-scale skill.

7 Conclusion

Among the evaluated systems, AI weather models do not show a uniform relative MAE deficit versus IFS in upper- and lower-tail regimes. On Track A, EPT-2.1 Europa leads all-conditions wind; EPT-2 HRRR has the highest temperature point estimate overall and in the heat regime; EPT-2.1 Helios leads solar overall and in the overcast and clear-sky tails; and three Jua systems gain about 14–15% at moderate precipitation. The counterexamples are equally important: AIFS has negative wind- and heat-tail skill, NOAA GFS has the largest heat-tail deficit, and EPT-2.1 Helios is the weakest AI system in the typical irradiance band.

Track B preserves the four-variable picture. EPT-2.1 Europa and ICON-EU are comparable in the gale and heat tails, EPT-2.1 Helios has the highest solar point estimate, and EPT-2 HRRR is the only regional system positive from light through high precipitation. At the heavy precipitation >P95 tail, effect sizes narrow but do not vanish: EPT-2 Reasoning remains modestly ahead, EPT-2.1 Europa is near IFS, and EPT-2 HRRR and EPT-2e trail it. More generally, every evaluated system overforecasts low observed values and underforecasts high ones after debiasing. The remaining failures at extremes are therefore model-specific against a shared centre-seeking bias, not a uniform property of AI weather models.

Data and code availability. Metrics were computed through a content-addressed station-verification API; requests and responses are archived. The reproduction package is at <https://github.com/juaAI/aiweather-extremes> and includes the aggregates used for the reported numbers, headline skill tables, and the extraction, aggregation and figure code. Score-based figures and tables regenerate offline from those aggregates; Figure 1 is an external illustrative asset and is not regenerated. Forecast fields used in the evaluation are available via the Jua platform (<https://jua.ai>).

References

- Zied Ben Bouallègue, Mariana C. A. Clare, Linus Magnusson, Estibaliz Gascón, Michael Maier-Gerber, Martin Janoušek, Mark Rodwell, Florian Pinault, Jesper S. Dramsch, Simon T. K. Lang, Baudouin Raoult, Florence Rabier, Matthieu Chevallier, Irina Sandu, Peter Dueben, Matthew Chantry, and Florian Pappenberger. The rise of data-driven weather forecasting: A first statistical assessment of machine learning-based weather forecasts in an operational-like context. *Bulletin of the American Meteorological Society*, 105(6):E864–E883, 2024. doi: 10.1175/BAMS-D-23-0162.1.
- Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian. Accurate medium-range global weather forecasting with 3D neural networks. *Nature*, 619(7970):533–538, 2023. doi: 10.1038/s41586-023-06185-3.
- Cristian Bodnar, Wessel P. Bruinsma, Ana Lucic, Megan Stanley, Anna Allen, Johannes Brandstetter, Patrick Garvan, Maik Riechert, Jonathan A. Weyn, Haiyu Dong, Jayesh K. Gupta, Kit Thambiratnam, Alexander T. Archibald, Chun-Chieh Wu, Elizabeth Heider, Max Welling, Richard E. Turner, and Paris Perdikaris. A foundation model for the Earth system. *Nature*, 641(8065):1180–1187, 2025. doi: 10.1038/s41586-025-09005-y.

- Massimo Bonavita. On some limitations of current machine learning weather prediction models. *Geophysical Research Letters*, 51(12):e2023GL107377, 2024. doi: 10.1029/2023GL107377.
- Andrew J. Charlton-Perez, Helen F. Dacre, Simon Driscoll, Suzanne L. Gray, Ben Harvey, Natalie J. Harvey, Kieran M. R. Hunt, Robert W. Lee, Ranjini Swaminathan, Remy Vandaele, and Ambrogio Volonté. Do AI models produce better weather forecasts than physics-based models? A quantitative evaluation case study of Storm Ciarán. *npj Climate and Atmospheric Science*, 7:93, 2024. doi: 10.1038/s41612-024-00638-w.
- Aman Gupta, Aditi Sheshadri, and Dhruv Suri. MAUSAM: An observations-focused assessment of global AI weather prediction models during the South Asian monsoon. arXiv preprint arXiv:2509.01879, 2025.
- Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, András Horányi, Joaquín Muñoz-Sabater, Julien Nicolas, Carole Peubey, Raluca Radu, Dinand Schepers, Adrian Simmons, Cornel Soci, Saleh Abdalla, Xavier Abellan, Gianpaolo Balsamo, Peter Bechtold, Gionata Biavati, Jean Bidlot, Massimo Bonavita, Giovanna De Chiara, Per Dahlgren, Dick Dee, Michail Diamantakis, Rossana Dragani, Johannes Flemming, Richard Forbes, Manuel Fuentes, Alan Geer, Leopold Haimberger, Sean Healy, Robin J. Hogan, Elías Hólm, Marta Janisková, Sarah Keeley, Patrick Laloyaux, Philippe Lopez, Cristina Lupu, Gabor Radnoti, Patricia de Rosnay, Iryna Rozum, Freja Vamborg, Sebastien Villaume, and Jean-Noël Thépaut. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.
- Weixin Jin, Jonathan Weyn, Pengcheng Zhao, Siqi Xiang, Jiang Bian, Zuliang Fang, Haiyu Dong, Hongyu Sun, Kit Thambiratnam, and Qi Zhang. WeatherReal: A benchmark based on in-situ observations for evaluating weather models. arXiv preprint arXiv:2409.09371, 2024.
- Jua AI. EPT-2 family: Models and products documentation. Jua documentation, <https://docs.jua.ai/models-and-products/jua-models/ept-2>, 2026. Accessed 18 July 2026; includes the EPT-2.1 Europa and EPT-2.1 Helios model specifications.
- Albert M. G. Klein Tank, Francis W. Zwiers, and Xuebin Zhang. Guidelines on analysis of extremes in a changing climate in support of informed decisions for adaptation. Technical Report WCDMP-No. 72, WMO-TD/No. 1500, World Meteorological Organization, Geneva, 2009.
- Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, Peter Wirnsberger, Meire Fortunato, Ferran Alet, Suman Ravuri, Timo Ewalds, Zach Eaton-Rosen, Weihua Hu, Alexander Meroze, Stephan Hoyer, George Holland, Oriol Vinyals, Jacklynn Stott, Alexander Pritzel, Shakir Mohamed, and Peter Battaglia. Learning skillful medium-range global weather forecasting. *Science*, 382(6677):1416–1421, 2023. doi: 10.1126/science.adi2336.
- Simon Lang, Mihai Alexe, Matthew Chantry, Jesper Dramsch, Florian Pinault, Baudouin Raoult, Mariana C. A. Clare, Christian Lessig, Michael Maier-Gerber, Linus Magnusson, Zied Ben Bouallègue, Ana Prieto Nemesio, Peter D. Dueben, Andrew Brown, Florian Pappenberger, and Florence Rabier. AIFS – ECMWF’s data-driven forecasting system. arXiv preprint arXiv:2406.01465, 2024.
- Roberto Molinaro. EPT-2.1 Helios: Europe’s most accurate solar model. Jua blog, <https://jua.ai/company/blog/ept2-helios>, July 2026. Published 15 July 2026; accessed 18 July 2026.
- Roberto Molinaro, Samuel Lanthaler, Bogdan Raonić, Tobias Rohner, Victor Armegioiu, Stephan Simonis, Dana Grund, Yannick Ramic, Zhong Yi Wan, Fei Sha, Siddhartha Mishra,

- and Leonardo Zepeda-Núñez. Generative AI for fast and accurate statistical computation of fluids. arXiv preprint arXiv:2409.18359, 2024.
- Roberto Molinaro, Niall Siegenheim, Niels Poulsen, Jordan Dane Daubinet, Henry Martin, Mark Frey, Kevin Thiart, Alexander Jakob Dautel, Andreas Schlueter, Alex Grigoryev, Bogdan Danciu, Nikoo Ekhtiari, Bas Steunebrink, Leonie Wagner, and Marvin Vincent Gabler. EPT-2 technical report. arXiv preprint arXiv:2507.09703, 2025.
- Roberto Molinaro, Niall Siegenheim, Henry Martin, Mark Frey, Niels Poulsen, Philipp Seitz, and Marvin Vincent Gabler. Universal diffusion-based probabilistic downscaling. arXiv preprint arXiv:2602.11893, 2026.
- Leonardo Olivetti and Gabriele Messori. Advances and prospects of deep learning for medium-range extreme weather forecasting. *Geoscientific Model Development*, 17(6):2347–2358, 2024a. doi: 10.5194/gmd-17-2347-2024.
- Leonardo Olivetti and Gabriele Messori. Do data-driven models beat numerical models in forecasting weather extremes? A comparison of IFS HRES, Pangu-Weather, and GraphCast. *Geoscientific Model Development*, 17(21):7915–7962, 2024b. doi: 10.5194/gmd-17-7915-2024.
- Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, Zongyi Li, Kamyar Azizzadenesheli, Pedram Hassanzadeh, Karthik Kashinath, and Animashree Anandkumar. FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators. arXiv preprint arXiv:2202.11214, 2022.
- Jaideep Pathak, Yair Cohen, Piyush Garg, Peter Harrington, Noah Brenowitz, Dale Durran, Morteza Mardani, Arash Vahdat, Shaoming Xu, Karthik Kashinath, and Michael Pritchard. Kilometer-scale convection-allowing model emulation using generative diffusion modeling. *Science Advances*, 12(5):eadv0423, 2026. doi: 10.1126/sciadv.adv0423.
- S. E. Perkins and L. V. Alexander. On the measurement of heat waves. *Journal of Climate*, 26(13):4500–4517, 2013. doi: 10.1175/JCLI-D-12-00383.1.
- Sarah E. Perkins. A review on the scientific understanding of heatwaves: Their measurement, driving mechanisms, and changes at the global scale. *Atmospheric Research*, 164–165:242–267, 2015. doi: 10.1016/j.atmosres.2015.05.014.
- Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. Probabilistic weather forecasting with machine learning. *Nature*, 637(8044): 84–90, 2025. doi: 10.1038/s41586-024-08252-9.
- Andreas Schlueter and Leonie Wagner. juaAI/stationbench: v0.1.1. Zenodo, <https://doi.org/10.5281/zenodo.14671604>, January 2025. Software, MIT license; <https://github.com/juaAI/stationbench>.
- Niall Siegenheim. EPT2-HRRR: The first operational high-resolution AI weather model for Europe. Jua blog, <https://jua.ai/company/blog/ept2-hrrr>, October 2025. Published 20 October 2025; accessed 18 July 2026.
- Christopher Subich, Syed Zahid Husain, Leo Separovic, and Jing Yang. Fixing the double penalty in data-driven weather forecasting through a modified spherical harmonic loss function. arXiv preprint arXiv:2501.19374, 2025. Accepted at the International Conference on Machine Learning (ICML) 2025.

- Xuebin Zhang, Lisa Alexander, Gabriele C. Hegerl, Philip Jones, Albert Klein Tank, Thomas C. Peterson, Blair Trewin, and Francis W. Zwiers. Indices for monitoring changes in extremes based on daily temperature and precipitation data. *WIREs Climate Change*, 2(6):851–870, 2011. doi: 10.1002/wcc.147.
- Zhongwei Zhang, Erich Fischer, Jakob Zscheischler, and Sebastian Engelke. Physics-based models outperform AI weather forecasts of record-breaking extremes. *Science Advances*, 12(18): eaec1433, 2026. doi: 10.1126/sciadv.aec1433.

A Supplementary material

Table S1: Track A MAE skill (%), 6–48 h, by regime – Figure 7, wind. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.4 (0.5)	-1.7 (1.4)	+6.4 (0.9)	+11.3 (0.7)	+8.5 (0.4)	+9.0 (1.2)
Jua EPT-2 HRRR	+4.5 (0.3)	+2.7 (1.0)	+1.4 (0.9)	+5.1 (0.3)	+4.9 (1.0)	+6.1 (0.9)
Jua EPT-2 Reasoning	+3.0 (0.1)	-0.7 (0.4)	+2.8 (0.5)	+5.6 (0.3)	+4.5 (0.3)	-0.7 (0.3)
Jua EPT-2e	-0.0 (0.2)	-9.1 (1.0)	-1.4 (1.1)	+5.6 (0.5)	+2.9 (0.6)	-6.6 (0.5)
Microsoft Aurora	-0.7 (0.3)	-6.6 (1.1)	-3.9 (1.1)	+1.9 (0.8)	+2.6 (0.8)	-3.4 (0.6)
ECMWF AIFS	-5.4 (0.4)	-9.8 (1.2)	-8.0 (1.3)	-2.1 (0.6)	-1.4 (0.8)	-10.6 (0.6)
ECMWF ENS (mean)	+1.4 (0.2)	-0.9 (0.7)	+1.5 (0.6)	+3.1 (0.4)	+2.1 (0.4)	-1.0 (0.2)
NOAA GFS	-11.7 (0.7)	+1.8 (1.4)	-9.5 (1.4)	-21.8 (1.4)	-16.3 (1.3)	+0.6 (0.7)
DWD ICON Global	+3.6 (0.4)	-4.1 (1.2)	+2.8 (1.0)	+2.6 (0.7)	+2.5 (0.7)	+9.5 (1.1)

Table S2: Track A MAE skill (%), 6–48 h – Figure 7, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+11.5 (0.7)	+4.7 (1.0)	+10.8 (1.3)	+11.0 (1.0)	+11.8 (1.1)	+15.0 (1.7)
Jua EPT-2 HRRR	+12.1 (0.4)	+9.5 (2.4)	+11.5 (1.2)	+9.4 (0.9)	+11.4 (1.0)	+19.6 (2.2)
Jua EPT-2 Reasoning	+4.6 (0.4)	-6.9 (1.3)	+2.6 (1.3)	+7.4 (0.4)	+7.0 (0.8)	+1.1 (0.7)
Jua EPT-2e	+0.8 (0.7)	-24.3 (3.6)	-5.0 (2.8)	+6.1 (0.8)	+7.4 (1.1)	-5.0 (0.7)
Microsoft Aurora	-1.3 (1.1)	-22.2 (4.4)	-7.0 (4.6)	+4.1 (1.0)	+3.8 (1.6)	-6.9 (1.0)
ECMWF AIFS	-1.4 (0.6)	-27.3 (3.6)	-5.1 (2.3)	+2.3 (1.0)	+4.9 (1.4)	-4.9 (2.0)
ECMWF ENS (mean)	-1.7 (0.3)	-12.2 (1.0)	-3.3 (0.9)	+1.4 (0.4)	+0.3 (0.8)	-6.1 (0.5)
NOAA GFS	-23.9 (1.4)	-25.3 (3.5)	-26.3 (2.9)	-26.5 (2.5)	-17.8 (2.1)	-22.8 (2.0)
DWD ICON Global	+5.5 (0.9)	-3.8 (1.2)	+3.0 (1.7)	+4.1 (1.5)	+8.3 (0.9)	+10.2 (2.0)

Table S3: Track A MAE skill (%), 1–48 h hourly – Figure 7, solar. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+10.2 (1.7)	+15.4 (5.0)	+16.4 (3.4)	-8.2 (1.1)	+13.7 (2.9)	+24.8 (5.4)
Jua EPT-2.1 Europa	+7.1 (1.0)	+3.1 (4.0)	+11.0 (3.1)	+9.2 (2.3)	+1.9 (4.3)	+4.4 (4.8)
Jua EPT-2 HRRR	+0.6 (1.4)	-3.6 (4.6)	+3.8 (3.7)	+4.5 (1.0)	-4.6 (4.8)	-4.1 (4.9)
Jua EPT-2 Reasoning	+6.3 (0.5)	+0.9 (1.3)	+9.5 (0.8)	+14.0 (0.7)	-0.8 (1.9)	-2.6 (2.6)
Jua EPT-2e	+2.0 (1.4)	-10.6 (3.3)	+5.9 (2.3)	+13.5 (1.6)	-7.5 (4.1)	-7.6 (6.0)
NOAA GFS	-11.2 (1.6)	-3.3 (4.4)	-24.3 (2.1)	-28.3 (2.5)	+8.8 (3.4)	+16.7 (4.2)
DWD ICON Global	+5.6 (0.8)	+6.5 (3.3)	+8.1 (2.6)	+3.1 (1.5)	+4.2 (3.3)	+6.9 (4.2)

Table S4: Track A MAE skill (%), 1–12 h – Figure 8, wind.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.0 (0.5)	-3.7 (1.0)	+3.6 (1.1)	+10.5 (0.7)	+8.1 (0.3)	+8.2 (0.9)
Jua EPT-2 HRRR	+2.0 (0.3)	+0.7 (1.0)	-2.1 (0.8)	+2.4 (0.2)	+3.0 (0.9)	+4.7 (0.9)
Jua EPT-2 Reasoning	+1.5 (0.1)	-1.5 (0.2)	+0.4 (0.3)	+3.2 (0.2)	+3.4 (0.2)	-0.4 (0.2)
Jua EPT-2e	-2.4 (0.2)	-10.8 (0.7)	-6.0 (0.7)	+2.1 (0.5)	+1.7 (0.4)	-6.2 (0.4)
ECMWF ENS (mean)	+0.4 (0.3)	–	–	–	–	–
NOAA GFS	-10.9 (0.8)	+4.8 (0.8)	-5.6 (0.9)	-21.5 (1.8)	-18.1 (1.4)	+0.7 (0.6)
DWD ICON Global	+3.7 (0.3)	-1.3 (0.6)	+4.0 (0.9)	+3.3 (0.7)	+1.5 (0.4)	+8.1 (1.1)

Table S5: Track A MAE skill (%), 1–12 h – Figure 8, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.3 (0.8)	+2.9 (1.1)	+7.3 (1.0)	+7.8 (1.1)	+7.5 (1.1)	+13.3 (1.4)
Jua EPT-2 HRRR	+9.5 (0.6)	+9.0 (2.3)	+8.7 (1.6)	+6.2 (1.0)	+8.2 (0.8)	+18.8 (2.0)
Jua EPT-2 Reasoning	+2.4 (0.2)	-5.3 (0.6)	+1.0 (0.6)	+4.3 (0.2)	+4.2 (0.5)	+0.0 (0.5)
Jua EPT-2e	-2.5 (0.6)	-24.3 (2.7)	-7.7 (2.2)	+2.3 (0.7)	+3.0 (1.1)	-7.2 (0.5)
ECMWF ENS (mean)	-3.0 (0.2)	–	–	–	–	–
NOAA GFS	-26.9 (1.5)	-29.6 (4.0)	-32.4 (3.5)	-29.0 (2.4)	-20.5 (1.6)	-23.9 (2.5)
DWD ICON Global	+5.3 (0.8)	-2.9 (1.0)	+2.8 (1.2)	+4.9 (1.5)	+6.3 (0.7)	+10.1 (1.4)

Table S6: Track A MAE skill (%), 1–12 h – Figure 8, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+13.5 (1.9)	+20.6 (5.9)	+20.1 (2.9)	-6.9 (1.5)	+17.7 (2.6)	+28.7 (6.1)
Jua EPT-2.1 Europa	+6.5 (1.4)	+1.2 (3.9)	+10.5 (3.8)	+8.1 (3.0)	+1.9 (4.3)	+5.6 (4.8)
Jua EPT-2 HRRR	-2.1 (1.8)	-6.8 (4.7)	+0.9 (4.3)	+0.3 (1.5)	-6.4 (5.0)	-3.7 (4.8)
Jua EPT-2 Reasoning	+4.4 (0.4)	+0.6 (1.3)	+7.4 (0.8)	+11.0 (0.7)	-2.4 (1.3)	-3.0 (1.7)
Jua EPT-2e	-1.1 (1.7)	-13.5 (3.4)	+4.1 (3.1)	+11.0 (1.7)	-13.8 (4.5)	-10.0 (5.0)
NOAA GFS	-12.7 (1.9)	-5.8 (4.2)	-27.6 (2.7)	-29.8 (2.6)	+9.7 (4.2)	+15.9 (3.6)
DWD ICON Global	+6.4 (1.2)	+6.2 (3.4)	+9.7 (3.0)	+3.4 (1.6)	+4.1 (2.9)	+8.8 (4.8)

Table S7: Track A precipitation MAE skill (%), 1–12 h (Figure 10). Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+1.9 (± 0.5)	-2.9 (± 1.0)	+3.3 (± 1.0)	+8.2 (± 1.0)	+7.8 (± 0.7)	+2.2 (± 0.5)
Jua EPT-2 HRRR	-2.7 (± 1.1)	-10.9 (± 3.8)	+3.9 (± 1.2)	+10.8 (± 1.5)	+6.8 (± 1.1)	-4.5 (± 0.8)
Jua EPT-2e	+4.6 (± 0.9)	+12.4 (± 3.4)	+8.9 (± 1.1)	+11.2 (± 1.1)	+2.2 (± 1.5)	-6.0 (± 1.3)
Jua EPT-2.1 Europa	+2.2 (± 1.5)	+22.0 (± 3.2)	-16.1 (± 4.2)	-16.5 (± 4.6)	-10.8 (± 3.2)	-1.1 (± 0.7)
NOAA GFS	-7.3 (± 1.0)	-3.4 (± 3.6)	-9.1 (± 1.4)	-14.2 (± 1.5)	-16.1 (± 1.9)	-4.8 (± 1.8)
DWD ICON Global	-1.4 (± 2.3)	+11.4 (± 3.3)	-17.4 (± 3.8)	-17.1 (± 4.5)	-11.5 (± 3.5)	-0.3 (± 0.6)

Table S8: Track A MAE skill (%) by lead – Figure 3, wind. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.3	+8.8	+8.7	+9.3	+8.5	+9.2	+8.1	+9.0
EPT-2 HRRR	+3.3	+5.0	+3.9	+5.6	+4.6	+6.0	+5.1	+6.9
EPT-2 Reasoning	+1.9	-0.7	+2.3	-0.8	+2.9	-0.9	+3.9	-0.4
EPT-2e	-1.2	-8.0	-1.0	-7.5	-0.1	-6.7	+1.1	-5.5
Aurora	-1.6	-4.0	-1.6	-3.6	-0.9	-3.6	+0.4	-2.8
AIFS	-6.6	-10.8	-6.5	-11.0	-5.7	-11.1	-4.0	-9.5
ENS	+0.3	-0.9	+0.7	-0.8	+1.2	-0.9	+2.3	-1.5
GFS	-11.1	+2.0	-11.1	+1.8	-11.8	+0.6	-12.3	-0.5
ICON Global	+4.6	+10.1	+4.6	+10.1	+4.0	+9.6	+2.4	+8.6

Table S9: Track A MAE skill (%) by lead – Figure 4, temperature. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.4	+12.5	+12.0	+15.8	+11.5	+14.5	+11.6	+14.4
EPT-2 HRRR	+10.1	+19.5	+12.6	+21.3	+12.1	+19.4	+12.2	+18.4
EPT-2 Reasoning	+2.2	-0.5	+3.4	+0.2	+4.5	+0.9	+6.0	+2.2
EPT-2e	-4.2	-10.0	-1.5	-7.5	+1.1	-4.7	+3.2	-2.7
Aurora	-4.6	-7.8	-2.3	-7.1	-1.2	-7.0	+0.2	-6.7
AIFS	-4.5	-7.5	-2.5	-5.2	-1.7	-5.0	+0.4	-3.7
ENS	-3.0	-6.7	-2.6	-6.3	-1.9	-6.0	-0.6	-5.6
GFS	-27.3	-24.6	-24.2	-22.5	-24.2	-23.2	-22.3	-22.6
ICON Global	+5.1	+10.1	+6.9	+11.7	+6.0	+10.2	+4.0	+8.8

Table S10: Track A MAE skill (%) by lead – Figure 5, solar (Ovc. = overcast, P5–25). Point estimates; SEs in regime tables.

Model	1 h		6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+26.1	–	+10.9	+19.0	+9.0	+15.2	+8.1	+15.4	+9.2	+15.5
EPT-2.1 Europa	+6.2	–	+5.6	+10.1	+6.7	+10.0	+6.1	+10.2	+6.3	+10.2
ICON Global	+7.1	–	+5.6	+9.8	+6.2	+8.9	+5.2	+8.1	+4.3	+6.3
EPT-2 Reasoning	+3.5	–	+4.2	+7.0	+5.1	+8.5	+6.0	+9.5	+7.8	+11.1
EPT-2e	+0.8	–	-4.5	+2.4	-2.5	+2.2	+0.2	+2.9	+4.1	+8.1
EPT-2 HRRR	-4.6	–	-4.8	-0.5	-2.7	+1.3	-1.7	+2.6	+1.1	+5.1
GFS	-12.8	–	-13.7	-30.2	-12.4	-29.4	-11.7	-25.8	-11.1	-24.3

Table S11: Track B wind MAE skill (%), 1–48 h, by regime (Figure 15).

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.2 (0.4)	+0.1 (0.8)	+3.0 (0.3)	+8.9 (0.6)	+8.2 (0.3)	+10.5 (1.4)
Jua EPT-2 HRRR	+2.4 (0.6)	+4.9 (1.0)	-1.8 (1.7)	+3.5 (0.4)	+1.8 (0.4)	+4.1 (1.5)
DWD ICON-EU	+5.3 (0.5)	-0.7 (1.0)	+1.8 (0.6)	+5.6 (0.5)	+3.9 (0.2)	+11.5 (1.6)

Table S12: Track B MAE skill (%), 1–48 h – Figure 15, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+13.2 (0.6)	+5.9 (1.8)	+11.7 (2.0)	+12.3 (1.3)	+13.5 (0.6)	+16.6 (2.3)
Jua EPT-2 HRRR	+12.0 (0.7)	+12.8 (3.3)	+14.9 (1.4)	+9.2 (0.4)	+8.7 (1.3)	+17.2 (3.5)
DWD ICON-EU	+12.4 (0.8)	+7.7 (1.1)	+11.8 (2.3)	+10.6 (1.1)	+11.3 (1.5)	+17.2 (3.2)

Table S13: Track B MAE skill (%), 1–48 h – Figure 15, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+12.6 (2.5)	+21.6 (4.6)	+18.1 (5.5)	-11.3 (0.5)	+13.5 (5.8)	+32.2 (4.9)
Jua EPT-2.1 Europa	+7.8 (2.0)	+7.4 (3.9)	+12.8 (3.1)	+10.7 (1.7)	+0.1 (4.5)	+3.4 (7.8)
Jua EPT-2 HRRR	+0.8 (2.2)	-0.2 (5.1)	+4.8 (5.7)	+6.5 (0.3)	-6.1 (8.2)	-6.4 (7.7)
DWD ICON-EU	+7.2 (1.4)	+12.4 (3.3)	+12.4 (2.5)	+1.0 (2.2)	+0.9 (3.3)	+10.5 (5.7)

Table S14: Track B MAE skill (%) by lead – Figure 17, wind.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.6	+11.1	+9.3	+12.0	+8.9	+11.9	+8.4	+12.3
EPT-2 HRRR	+2.3	+3.8	+3.3	+5.0	+3.9	+5.4	+4.7	+6.7
ICON-EU	+7.5	+14.6	+7.7	+14.7	+7.1	+13.9	+5.4	+13.7

Table S15: Track B MAE skill (%) by lead – Figure 17, temperature.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+10.7	+12.7	+14.6	+17.0	+13.2	+15.1	+13.5	+16.1
EPT-2 HRRR	+12.0	+17.1	+14.2	+19.3	+12.4	+16.8	+12.2	+16.9
ICON-EU	+13.7	+18.4	+14.9	+19.9	+13.2	+17.2	+10.6	+15.5

Table S16: Track B MAE skill (%) by lead – Figure 17, solar (Ovc. = overcast, P5–25).

Model	6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+12.9	+20.3	+11.4	+17.1	+10.4	+16.4	+11.2	+16.8
EPT-2.1 Europa	+7.2	+13.6	+8.0	+13.5	+6.8	+12.3	+6.9	+11.9
EPT-2 HRRR	-3.5	+3.9	-1.5	+4.7	-1.3	+4.6	+1.1	+6.4
ICON-EU	+7.4	+14.1	+7.8	+13.4	+6.0	+12.0	+4.4	+9.3

Table S17: Track B precipitation MAE skill (%) by lead (Figure 14). Point estimates.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
Jua EPT-2.1 Europa	+1.9	+0.5	+9.9	+1.0	+9.0	+1.6	+7.7	+1.9
Jua EPT-2 HRRR	-2.5	-0.6	-2.6	+0.6	-0.9	+1.0	+0.4	+1.7
DWD ICON-EU	-0.1	+0.4	+8.7	-0.6	+7.1	-0.1	+3.2	-1.2

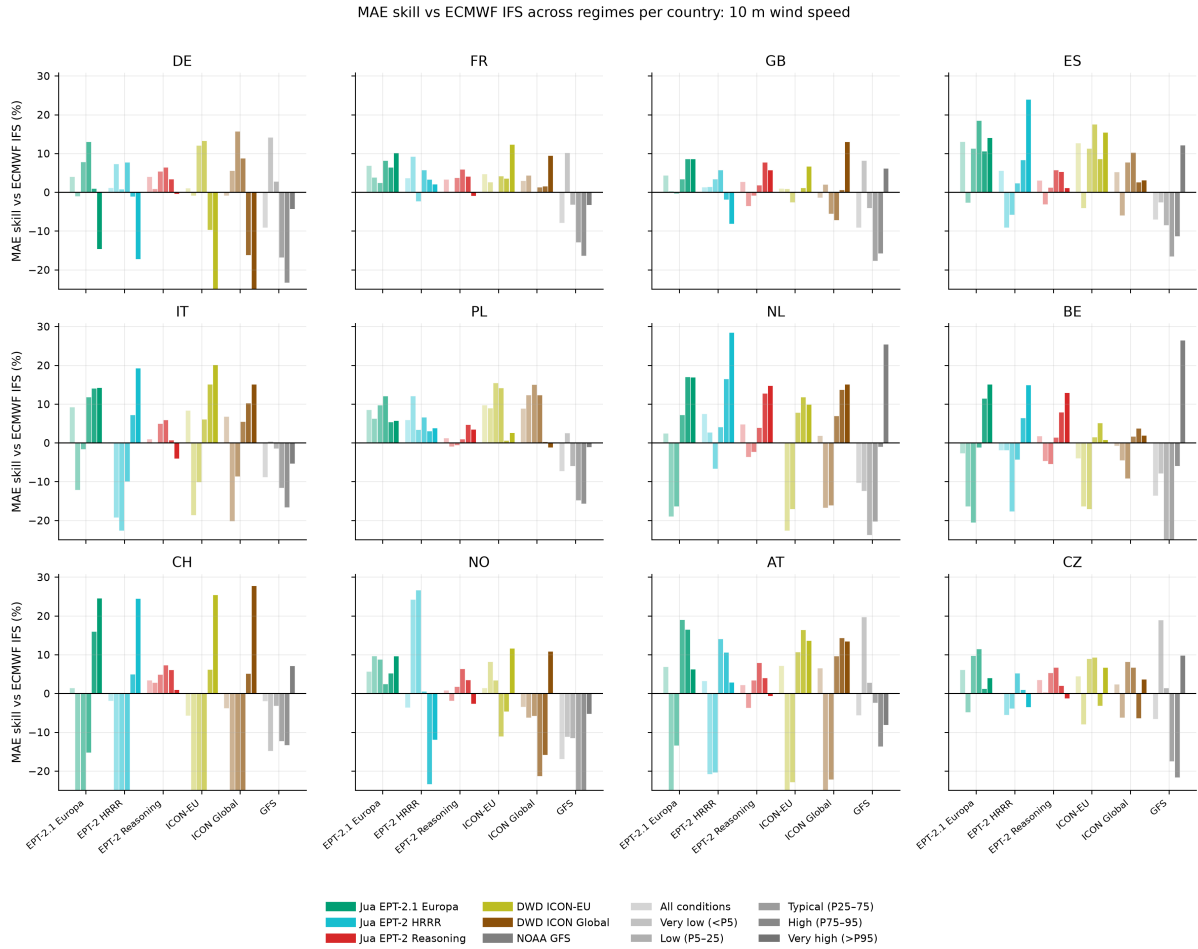


Figure S1: Per-country MAE skill by ERA5 1991–2020 climatological regime, 10 m wind, March–June 2026, pooled 1 to 48 h (hourly). Mixed regional/global model set (Section 4.5).

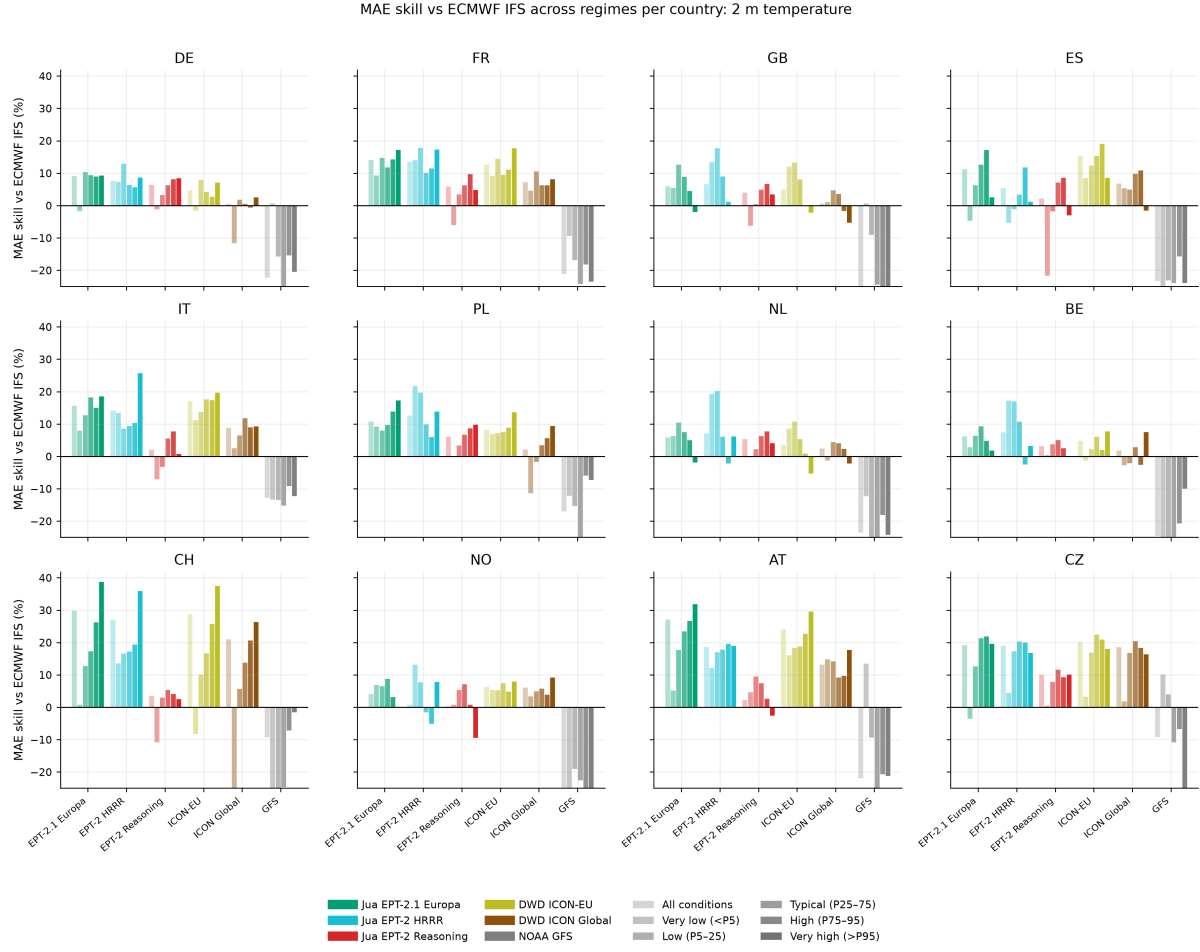


Figure S2: As Figure S1, for 2 m temperature.

Table S18: Track B MAE skill (%) for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased – Figure 13. Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.8 (± 0.9)	+29.3 (± 1.4)	-19.1 (± 6.0)	-20.2 (± 7.2)	-9.0 (± 5.7)	+1.2 (± 0.8)
Jua EPT-2 HRRR	+1.4 (± 3.4)	-3.6 (± 9.7)	+10.3 (± 4.4)	+15.6 (± 6.1)	+10.4 (± 2.5)	-0.7 (± 0.3)
DWD ICON-EU	+3.5 (± 0.5)	+15.3 (± 0.3)	-10.9 (± 2.7)	-11.4 (± 3.8)	-6.9 (± 2.2)	+0.1 (± 0.5)

Table S19: Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased – Figure 11. POD: probability of detection; FAR: false-alarm ratio; CSI: critical success index; frequency bias is forecast over observed event count.

Model	POD	FAR	CSI	Freq. bias
Jua EPT-2.1 Europa	0.28	0.62	0.19	0.74
DWD ICON Global	0.29	0.65	0.19	0.82
Jua EPT-2 Reasoning	0.23	0.56	0.18	0.53
ECMWF IFS	0.25	0.65	0.17	0.71
NOAA GFS	0.25	0.66	0.17	0.74
Jua EPT-2 HRRR	0.20	0.53	0.16	0.42
Jua EPT-2e	0.20	0.55	0.16	0.44
ECMWF ENS (mean)	0.15	0.50	0.13	0.29

MAE skill vs ECMWF IFS across regimes per country: 1 h shortwave accumulation

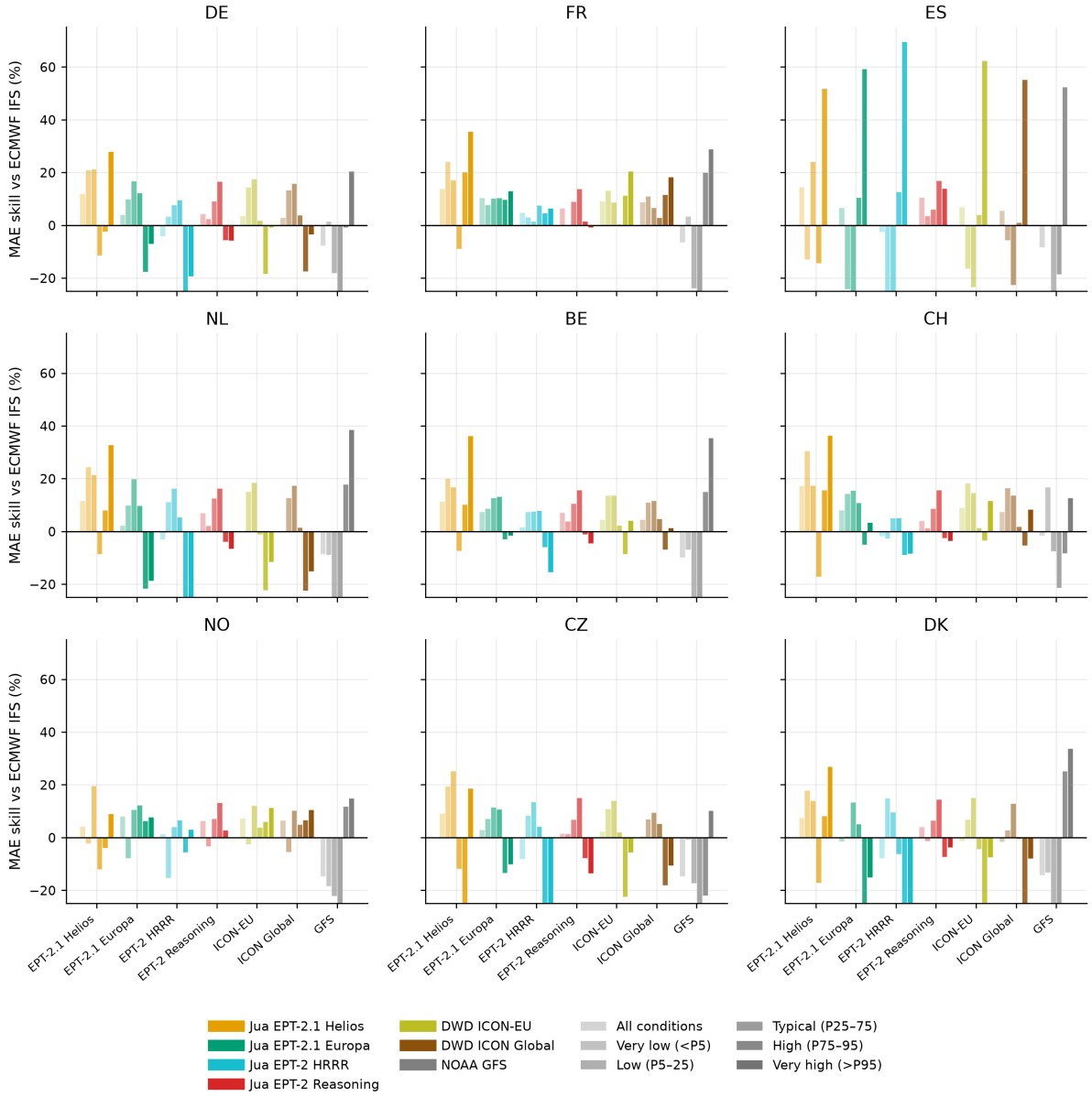


Figure S3: As Figure S1, for 1 h shortwave, with EPT-2.1 Helios added. GB, IT and PL have no solar station observations (Section 3.2).

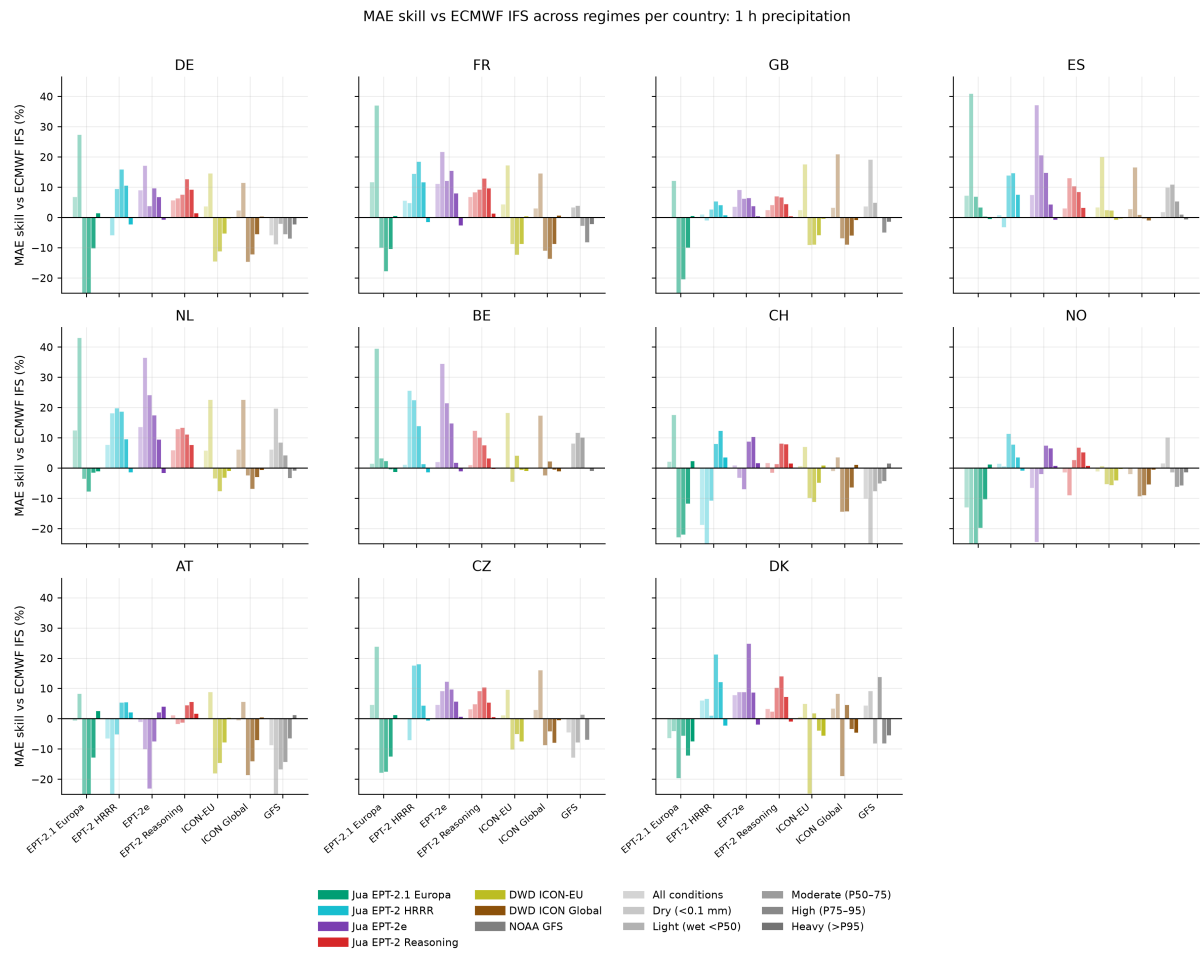


Figure S4: As Figure S1, for 1 h precipitation using the wet-hour regimes of Table 3. IT and PL have no gauges in the precipitation panel.